

Comparing Mutual Information in Finite Samples: Wald and Expanded Welch–Satterthwaite Inference

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Abstract

Mutual information (MI) measures association between two variables. This thesis asks whether two independently sampled populations have equal MI, even when their joint probability tables differ. The usual Normal Wald test uses a normal approximation that can be inaccurate in small or sparse samples.

The thesis develops an Expanded Welch–Satterthwaite adjustment for this comparison. It derives how changing one observation affects the estimated MI variance, including changes to the cell probabilities and their row and column totals. That calculation determines the degrees of freedom for a Student t reference. Expanded Welch retains Wald’s estimated MI difference and standard error; it changes only the distribution used to judge the resulting statistic.

The methods are evaluated in 3,111 fixed statistical settings, each with 20,000 independently sampled table pairs. The design includes square tables from 2×2 to 8×8 , selected rectangular tables, unequal sample sizes, sparse cells, and populations with different distributions but equal MI. At a nominal 5% threshold, the experiment measures false positives, detection when MI differs, and the frequency of usable results.

Expanded Welch reduces excessive Wald false positives in some settings. However, when both tests return a p-value, its Student reference gives a p-value no smaller than Wald’s. It therefore also lowers rejection where Wald already rejects too rarely and can reduce detection when MI differs. In sparse samples, Expanded Welch more often fails to return a usable p-value under the implemented rule. Wald is closer to the target 5% rate in all 24 main null settings with sample sizes 500 or 1,000. A supplementary diagnostic shows that changing the reference distribution cannot correct error in the standard error shared by both tests. For the settings studied, Wald remains the simpler general baseline. Expanded Welch is a potential additional check when a separate, closely matched calibration study shows that Wald rejects equal MI too often.

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Chapter 1

Introduction

Suppose two independent groups are observed on the same categorical variables X and Y . Mutual information (MI) describes the strength of association between those variables in each group. If the estimated MI differs, the difference alone does not tell us whether the population MI values differ: both estimates vary when new observations are drawn. The question in this thesis is whether the two *population* MI values can plausibly be equal.

This is not the usual test of whether X and Y are independent within one group. The null hypothesis here permits dependence in both groups and permits their complete joint distributions to differ. It specifies only

$$H_0 : I(P) = I(Q), \quad H_1 : I(P) \neq I(Q), \quad (1.1)$$

where P and Q are the two joint distributions. The samples from P and Q are independent of each other; that sampling assumption does not imply independence of X and Y within either population.

1.1 Why the comparison is difficult

MI uses the probabilities in every cell of a table and the totals of its rows and columns. In a small sample, some cells can be empty even when their population probabilities are positive. The observed tables determine both the estimated MI difference and its standard error, so sparse data can affect both. The Normal Wald test compares their ratio with a normal curve, but in small samples it may reject equal MI more or less often than the intended 5%.

Expanded Welch offers a possible adjustment. It uses the same estimated difference and standard error as Wald, but compares their ratio with a Student t

distribution. Its degrees of freedom depend on how much the estimated MI variance changes from sample to sample. To calculate that change, we must account for the fact that changing one cell also changes its row and column totals and the MI values calculated from them. This thesis derives that calculation and tests whether the adjustment helps.

The adjustment is not automatically better. It can reduce false positives where Wald rejects too often, but it can also reduce detection where Wald already rejects too rarely. In sparse samples, it may fail to return a usable p-value when Wald succeeds. The experiment therefore examines false positives, detection and the frequency of usable results together.

1.2 Research questions

The thesis asks two main questions. First, how can changes in the estimated MI variance be used to choose the degrees of freedom for a Student test? Second, does the resulting Expanded Welch test offer a useful improvement over Normal Wald? Answering the second question requires looking at false positives, detection when MI differs, and the frequency of usable results. The experiment checks how these outcomes change with table size, uneven cell probabilities, the amount and location of association, unequal sample sizes and larger samples. It also measures the extra computation required.

1.3 Contributions to the field

This thesis combines established ideas about MI and Welch–Satterthwaite tests for a specific question: whether two populations have equal MI. Its contributions are the calculation needed for that test and evidence about whether the test is useful.

1. **Mathematical contribution.** The thesis derives how changing one observation affects the estimated MI variance. The calculation includes the resulting changes in the row and column totals and in pointwise MI.
2. **Methodological contribution.** Expanded Welch uses this calculation to choose the degrees of freedom for a two-sample Student test. The main experiment compares it with Wald; a supplementary analysis checks simpler sample-size choices for the degrees of freedom.
3. **Empirical contribution.** A reproducible experiment measures false positives, detection, usable results and runtime on fixed populations with known MI. It

covers 2×2 through 8×8 square tables, rectangular tables, unequal samples and very sparse settings.

4. **Practical contribution.** Expanded Welch can help when Wald rejects equal MI too often. It can also make an already-low rejection rate lower and return fewer usable results in sparse samples. The evidence shows why it should not replace Wald in general and points to problems that changing the reference distribution alone cannot solve.

1.4 Scope and organisation

The analysis concerns independent samples recorded in count tables with the same row and column categories. The main derivation assumes every population cell has positive probability and that its leading large-sample MI variance is greater than zero. Exact independence is considered separately because that variance becomes zero. The experiment compares Normal Wald with Expanded Welch using a two-sided 5% test. It does not establish performance for paired data, continuous MI estimators or a number of categories that grows with sample size.

Chapter 2 supplies the statistical background. Chapters 3 and 4 define the shared statistic and derive the correction. Chapter 5 explains how the populations and samples are generated and how the results are measured. Chapter 6 presents and interprets the main comparisons in sequence. Chapter 7 draws out their broader implications and limits, and Chapter 8 states the contribution and recommendation. The appendices give the detailed algebra, population settings, supporting figures and source-to-result record.

Chapter 2

Background and Literature Review

This chapter develops the concepts needed to compare MI between two categorical populations. It begins with entropy, pointwise MI, and population MI. It then reviews plug-in estimation, first-order inference, the Welch–Satterthwaite approximation, and prior comparisons of information quantities. The final section identifies the gap addressed by the expanded method.

2.1 Information-theoretic foundations

2.1.1 Entropy and information content

For a discrete random variable X with probability mass function $p_X(x)$, the information content of outcome x is $-\log p_X(x)$. Rare outcomes have larger information content because they are less expected. Using natural logarithms gives information in nats. The Shannon entropy is the expected information content,

$$H(X) = - \sum_x p_X(x) \log p_X(x). \quad (2.1)$$

Entropy summarises the uncertainty of one random variable before its value is observed ([Shannon, 1948](#)).

For a pair (X, Y) , the joint entropy $H(X, Y)$ is defined from the joint probabilities in the same way. The conditional entropy $H(X | Y) = H(X, Y) - H(Y)$ measures the uncertainty remaining in X after observing Y .

2.1.2 Pointwise and average mutual information

Let $P = (p_{ij})$ be the joint distribution of categorical variables $X \in \{1, \dots, r\}$ and $Y \in \{1, \dots, c\}$. Write

$$p_{i+} = \sum_{j=1}^c p_{ij}, \quad p_{+j} = \sum_{i=1}^r p_{ij} \quad (2.2)$$

for the row and column margins. The pointwise mutual information associated with cell (i, j) is

$$\ell_P(i, j) = \log \left(\frac{p_{ij}}{p_{i+}p_{+j}} \right). \quad (2.3)$$

The denominator is the cell probability under independence with the same margins. A positive value means that the cell occurs more often than independence predicts; a negative value means that it occurs less often.

Mutual information is the probability-weighted average of these pointwise values (Shannon, 1948; Bossomaier et al., 2016):

$$I(P) = \sum_{i=1}^r \sum_{j=1}^c p_{ij} \ell_P(i, j). \quad (2.4)$$

Here the summations are nested: for each row i , the inner sum runs over all columns j . Equivalently, the expression sums once over every cell in the table and may be written with the shorthand $\sum_{i,j}$. Because P is the joint distribution of (X, Y) , the same quantity is conventionally written as $I(X; Y)$. Equivalently,

$$I(X; Y) = H(X) + H(Y) - H(X, Y). \quad (2.5)$$

The quantity is nonnegative and equals zero exactly when $p_{ij} = p_{i+}p_{+j}$ for every cell, including zero-probability cells (Bossomaier et al., 2016).

MI measures the overall departure from independence, not one particular shape of association.

This scalar summary deliberately discards much of the information in the joint distribution. Two populations can have the same MI while assigning probability to different cells, having different margins, or exhibiting different association patterns. Equality of MI is therefore weaker than equality of distributions. Our null requires only this one number to agree, not that the two tables match cell by cell.

The logarithm base determines the reporting unit. This thesis uses natural logarithms, so MI is measured in nats; dividing by $\log 2$ converts to bits. This conversion rescales MI and its standard error together but leaves the degrees of freedom unchanged, so the test result is identical in either unit.

2.2 Estimating discrete mutual information

Suppose a sample of size n produces counts N_{ij} . The empirical cell probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}}{n}. \quad (2.6)$$

Substitution into Equation (2.4) gives the plug-in estimator

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right). \quad (2.7)$$

The zero-cell convention follows from $\lim_{x \downarrow 0} x \log x = 0$. Empty cells contribute no direct term, but they still affect the empirical margins and the finite-sample behaviour of the estimator.

Under multinomial sampling, the cell counts are random but constrained to sum to n , so they are dependent. The plug-in estimator remains consistent for a fixed alphabet, but its finite-sample behaviour depends on the complete probability table. In particular, rare cells can change the sampling behaviour even when the alphabet and sample sizes are unchanged (Paninski, 2003).

For a fixed positive $r \times c$ support, the plug-in estimator has a positive leading bias as the sample size grows:

$$\text{Bias}\{\hat{I}(P)\} = \frac{(r-1)(c-1)}{2n} + O(n^{-2}). \quad (2.8)$$

The leading term grows with table size and shrinks as the sample grows. It does not describe every small or sparse sample. The result is closely related to Miller's entropy correction (Miller, 1955), and the same histogram-MI term appears in Moddemeijer (1989). Sparse sampling can produce further error beyond this leading approximation (Paninski, 2003).

Bias matters directly when comparing two populations. If $I(P) = I(Q)$ but $n_P \neq n_Q$, the uncorrected difference has leading bias

$$\frac{(r-1)(c-1)}{2} \left(\frac{1}{n_P} - \frac{1}{n_Q} \right). \quad (2.9)$$

For a 2×2 table with $n_P = 50$ and $n_Q = 100$, Equation (2.9) gives 0.005 nats despite equal population MIs. Chapter 3 corrects this sample-size effect by subtracting $(r-1)(c-1)/(2n)$ from each estimate.

2.3 Sampling variance of plug-in MI

To compare two estimated MI values, it is necessary to know how much each estimate would vary across repeated samples. The goal of this section is to derive the large-sample approximation

$$\text{Var}\{\hat{I}(P)\} \approx \frac{V(P)}{n}. \quad (2.10)$$

Here $V(P)$, defined below, measures how much the contribution of one observation varies across the population. The derivation assumes that the population cell probabilities are positive, so the logarithms and their derivatives are well defined. This first-order variance is established for histogram and discrete MI (Moddemeijer, 1989; Brillinger, 2004). The argument proceeds in three steps. First, sampling changes the empirical cell probabilities. Second, a first-order Taylor expansion translates those probability errors into an MI error. Third, that error can be written as a sample average, whose variance is straightforward to calculate.

Step 1: From probability errors to MI error.

Using the entropy identity in Equation (2.5), MI can be written directly in terms of the joint and marginal probabilities:

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (2.11)$$

Let $f(u) = u \log u$. The derivative needed for its first-order Taylor expansion follows directly from the product rule:

$$f'(u) = \frac{d}{du} \{u \log u\} = \log u + u \frac{1}{u} = \log u + 1. \quad (2.12)$$

The general first-order Taylor formula is

$$f(x) \approx f(a) + f'(a)(x - a), \quad (2.13)$$

where a is the value around which the function is expanded and $x - a$ is the distance from that value. Here the new value is the estimated probability $x = \hat{u}$, the expansion point is the population probability $a = u$, and $f(t) = t \log t$. Therefore, $x - a = \hat{u} - u$, which is the sampling error in the estimated probability. Substituting these quantities into the general formula gives

$$\hat{u} \log \hat{u} \approx \underbrace{u \log u}_{\text{population value}} + \underbrace{(\log u + 1)(\hat{u} - u)}_{\text{first-order change}}. \quad (2.14)$$

Subtracting the population value from both sides makes the change explicit:

$$\hat{u} \log \hat{u} - u \log u \approx (\log u + 1)(\hat{u} - u). \quad (2.15)$$

Applying this expansion to the three sums in Equation (2.11) gives

$$\begin{aligned} \hat{I}(P) - I(P) &\approx \sum_{i,j} (\log p_{ij} + 1)(\hat{p}_{ij} - p_{ij}) \\ &\quad - \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) \\ &\quad - \sum_j (\log p_{+j} + 1)(\hat{p}_{+j} - p_{+j}). \end{aligned} \quad (2.16)$$

The margin errors are sums of the corresponding cell errors:

$$\hat{p}_{i+} - p_{i+} = \sum_j (\hat{p}_{ij} - p_{ij}), \quad \hat{p}_{+j} - p_{+j} = \sum_i (\hat{p}_{ij} - p_{ij}). \quad (2.17)$$

For example, the row-margin sum can now be written as

$$\begin{aligned} \sum_i (\log p_{i+} + 1)(\hat{p}_{i+} - p_{i+}) &= \sum_i \left[(\log p_{i+} + 1) \left\{ \sum_j (\hat{p}_{ij} - p_{ij}) \right\} \right] \\ &= \sum_{i,j} (\log p_{i+} + 1)(\hat{p}_{ij} - p_{ij}). \end{aligned} \quad (2.18)$$

The inner sum over j is the probability error for a fixed row i . It is multiplied by that row's coefficient $\log p_{i+} + 1$; the outer sum then adds these products over all rows. The column-margin sum can be rearranged in the same way. Consequently, the same cell error $\hat{p}_{ij} - p_{ij}$ appears in the joint term, its row-margin term and its column-margin term. Its combined coefficient is

$$\begin{aligned} &(\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \\ &= \log \left(\frac{p_{ij}}{p_{i+} p_{+j}} \right) - 1 \\ &= \ell_P(i, j) - 1. \end{aligned} \quad (2.19)$$

Therefore,

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \{\ell_P(i, j) - 1\}(\hat{p}_{ij} - p_{ij}). \quad (2.20)$$

The empirical and population cell probabilities both sum to one, so

$$\sum_{i,j} (\hat{p}_{ij} - p_{ij}) = \sum_{i,j} \hat{p}_{ij} - \sum_{i,j} p_{ij} = 1 - 1 = 0. \quad (2.21)$$

The constant -1 in Equation (2.19) therefore contributes zero. Explicitly,

$$\begin{aligned}
 \hat{I}(P) - I(P) &\approx \sum_{i,j} \{\ell_P(i,j) - 1\}(\hat{p}_{ij} - p_{ij}) \\
 &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}) - \sum_{i,j} (\hat{p}_{ij} - p_{ij}) \\
 &= \sum_{i,j} \ell_P(i,j)(\hat{p}_{ij} - p_{ij}). \tag{2.22}
 \end{aligned}$$

The complicated changes in the joint table and its two margins have therefore reduced to a weighted sum of cell-probability errors. The weight for a cell is its population pointwise MI.

Step 2: From cell errors to observation-level errors.

Let $Z_a = (X_a, Y_a)$ be observation a . An empirical cell probability is the proportion of observations in that cell, so

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\}. \tag{2.23}$$

Substituting this expression into Equation (2.22) first gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i,j) \left[\frac{1}{n} \sum_{a=1}^n \{\mathbf{1}(Z_a = (i,j)) - p_{ij}\} \right]. \tag{2.24}$$

The finite sums can be exchanged, so the contribution from observation a is gathered inside one pair of brackets:

$$\begin{aligned}
 \hat{I}(P) - I(P) &\approx \frac{1}{n} \sum_{a=1}^n \left[\sum_{i,j} \ell_P(i,j) \mathbf{1}(Z_a = (i,j)) - \sum_{i,j} p_{ij} \ell_P(i,j) \right] \\
 &= \frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}. \tag{2.25}
 \end{aligned}$$

The first sum inside the brackets selects the single cell containing Z_a , giving $\ell_P(Z_a)$. The second is the probability-weighted average of all pointwise-MI values, which is $I(P)$. This equation is the key step. To first order, the error in the plug-in MI estimate behaves like the average of n observation-level contributions. The contribution of an observation in cell (i,j) is simply that cell's pointwise MI minus the population average, $\ell_P(i,j) - I(P)$. Its mean is zero because

$$E_P\{\ell_P(X,Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i,j) - I(P)\}$$

$$= \underbrace{\sum_{i,j} p_{ij} \ell_P(i, j)}_{I(P)} - I(P) \underbrace{\sum_{i,j} p_{ij}}_1 = 0. \quad (2.26)$$

In formal asymptotic terminology, the centred function $\ell_P(i, j) - I(P)$ is the influence function of MI; Equation (2.25) is its first-order linear representation (Hampel, 1974; van der Vaart, 1998; Kandasamy et al., 2015).

Step 3: Calculate the sampling variance.

Define the population variance of these centred contributions by

$$V(P) = \text{Var}_P\{\ell_P(X, Y) - I(P)\} = \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2. \quad (2.27)$$

Under independent sampling, the n terms in Equation (2.25) are independent and each has variance $V(P)$. The covariance between any two different observations is therefore zero. Hence

$$\begin{aligned} \text{Var}\{\hat{I}(P)\} &\approx \text{Var}\left\{\frac{1}{n} \sum_{a=1}^n \{\ell_P(Z_a) - I(P)\}\right\} \\ &= \frac{1}{n^2} \sum_{a=1}^n \text{Var}\{\ell_P(Z_a) - I(P)\} \\ &= \frac{1}{n^2} \sum_{a=1}^n V(P) = \frac{nV(P)}{n^2} = \frac{V(P)}{n}. \end{aligned} \quad (2.28)$$

This proves the approximation stated in Equation (2.10). It is the familiar result that the variance of an average of n independent contributions is the variance of one contribution divided by n .

The independence of observations is needed in the second line of Equation (2.28): it makes covariance terms between different observations vanish. Moddemeijer (1999) studies MI sampling variance when observation pairs are dependent and those terms must instead be accounted for. The present test and experiment concern independent multinomial samples, so they use $V(P)/n$ without a serial-dependence correction.

When $V(P) > 0$, the central limit theorem also gives

$$\frac{\hat{I}(P) - I(P)}{\sqrt{V(P)/n}} \xrightarrow{d} N(0, 1). \quad (2.29)$$

This does not claim that every finite-sample MI estimate is normally distributed. It is a local, large-sample approximation for a fixed table with positive population cell probabilities. The $1/n$ leading bias from Section 2.2 decreases faster than the $1/\sqrt{n}$ sampling error, so it does not change this limiting distribution.

The independence boundary.

At exact independence, $p_{ij} = p_{i+}p_{+j}$, so $\ell_P(i, j) = \log 1 = 0$ in every cell. Hence $I(P) = 0$, $V(P) = 0$, and the first-order average in Equation (2.25) vanishes.

A finite sample can still have positive MI. Its leading variation comes from the squared cell-probability errors in a second-order Taylor expansion, which produces the usual chi-squared limit (Brillinger, 2004; Marinescu and Balcau, 2025). The normal limit for comparing positive MI values and the chi-squared limit for testing independence therefore apply in different settings.

2.4 Approaches to Testing MI Differences

The methods below answer related but different questions. Some quantify the sampling error of one MI estimate; others compare two estimates or adjust a test when its standard error is itself estimated. For this thesis, a useful predecessor must be assessed against the null $I(P) = I(Q)$, which allows the two population tables to differ.

Normal Wald inference.

A Wald test estimates a parameter difference, divides it by an estimated standard error, and compares the result with its asymptotic normal reference (Wald, 1943; van der Vaart, 1998). For two independent MI estimates, the two first-order variance contributions add:

$$\text{Var}\{\hat{I}(P) - \hat{I}(Q)\} \approx \frac{V(P)}{n_P} + \frac{V(Q)}{n_Q}. \quad (2.30)$$

Each term is the sampling variance of one MI estimate. Independence of the two samples makes their covariance zero. Replacing the population variances by estimates gives the standardisation

$$\frac{\hat{I}(P) - \hat{I}(Q)}{\sqrt{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q}} \xrightarrow{d} N(0, 1) \quad \text{under } I(P) = I(Q). \quad (2.31)$$

Here $\hat{V}(P)$ and $\hat{V}(Q)$ estimate the pointwise-MI variances $V(P)$ and $V(Q)$. This limit assumes the regular positive-support conditions of Section 2.3 and increasing sample sizes. Chapter 3 applies the leading bias correction to the numerator and specifies the full Normal Wald test.

With smaller samples, the MI difference may not follow the normal approximation closely, and the estimated standard error can vary between samples. The Wald test still uses a normal reference; it makes no separate small-sample adjustment for that

standard-error variability.

The Welch–Satterthwaite principle.

Welch’s test compares two means when their population variances may differ (Welch, 1947). For two independent samples with means $\bar{X}_P, \bar{X}_Q$ and variances s_P^2, s_Q^2 , the statistic for testing equal population means and its effective degrees of freedom are

$$T_{\text{Welch}} = \frac{\bar{X}_P - \bar{X}_Q}{\sqrt{s_P^2/n_P + s_Q^2/n_Q}}, \quad (2.32)$$

$$\nu = \frac{(s_P^2/n_P + s_Q^2/n_Q)^2}{\frac{(s_P^2/n_P)^2}{n_P - 1} + \frac{(s_Q^2/n_Q)^2}{n_Q - 1}}. \quad (2.33)$$

The denominator of the statistic contains one estimated variance contribution from each sample. Under normal sampling, their component degrees of freedom are $n_P - 1$ and $n_Q - 1$. If s_P^2/n_P dominates, the formula makes the combined ν close to $n_P - 1$; when both contributions matter, both component values affect ν . Satterthwaite obtains this form by matching the mean and variance of their sum to a scaled chi-squared variable (Satterthwaite, 1946).

Welch compares the statistic with a Student distribution using the calculated ν . At the same significance level, this requires a larger absolute statistic to reject the null than a standard normal reference does. The difference between the Student and normal rejection thresholds becomes smaller as ν increases.

For MI, the estimated variance contributions are $\hat{V}(P)/n_P$ and $\hat{V}(Q)/n_Q$. Their component degrees of freedom need their own calculation because $\hat{V}(P)$ and $\hat{V}(Q)$ are nonlinear functions of the probability tables. Chapter 4 derives these MI-specific inputs to the same moment-matching structure.

Regression studies have also adjusted reference distributions to account for uncertainty in estimated variances (Pan and Wall, 2002; Bell and McCaffrey, 2002; Kauermann and Carroll, 2001; Fay and Graubard, 2001; Imbens and Kolesár, 2016). For example, Pan and Wall (2002) calculate approximate degrees of freedom for inference using a sandwich variance estimate. This supports the idea of examining the variance estimator itself, but the MI variance is a different function of the observed table and needs its own calculation. Bell and McCaffrey (2002) report regression variance estimates biased downward, while Kauermann and Carroll (2001) examine intervals that cover too rarely. A more cautious reference can help in such settings. In the MI comparison examined in Section 6.5, the average squared plug-in standard error exceeds the measured finite-sample variance of the estimated

difference. A Student reference cannot correct that discrepancy and further lowers rejection in that setting.

Prior MI inference and related comparisons.

MI uncertainty also has a broader literature than its leading normal approximation. [Roulston \(1999\)](#) derives entropy and MI error estimates using error propagation and examines them numerically. [Hutter \(2002\)](#) and [Hutter and Zaffalon \(2005\)](#) derive moments of the posterior distribution of MI under a Dirichlet model, including higher-order variance and shape information. Their analysis describes posterior uncertainty under a Dirichlet model, while this thesis evaluates how a two-sample test behaves over repeated samples. These results therefore answer a different uncertainty question from the Student reference used here.

Hutcheson’s test is a direct information-theoretic Welch predecessor. It compares two Shannon diversity estimates with an estimated variance and a Student reference, using

$$\nu_H = \frac{(s_{H,P}^2 + s_{H,Q}^2)^2}{(s_{H,P}^2)^2/n_P + (s_{H,Q}^2)^2/n_Q} \quad (2.34)$$

as its effective degrees of freedom ([Hutcheson, 1970](#)). Here each of $s_{H,P}^2$ and $s_{H,Q}^2$ is the estimated sampling variance of the corresponding entropy estimator $\widehat{H}$, and each variance component is assigned its sample size, not $n_i - 1$. Applying the same assignment to the two MI variance contributions gives a clear historical baseline. It does not, however, derive how the plug-in MI variance itself fluctuates: MI depends on a joint distribution and both margins, so moving mass into one cell changes pointwise values elsewhere in its row and column. The expanded calculation retains these changes.

The question of comparing MI values also has predecessors. In a segregation application, [Mora and Ruiz-Castillo \(2009\)](#) develop inference for an MI index and discuss pairwise comparisons across groups and periods; the later peer-reviewed account places that MI index within the wider segregation literature ([Mora and Ruiz-Castillo, 2011](#)). They do not derive the variance-sensitivity degrees of freedom used here. [Guha and Chothia \(2014\)](#) use MI for a different two-sample question: whether two continuous distributions are identical. Their MI is between a pooled observation and its sample label; their null is equality of distributions, not equality of two within-population MI values. For discrete data, [Drake and Guha \(2014\)](#) develop an MI-based k -sample test for the same distributional-equality question.

At exact independence, the first-order sampling variation of MI vanishes and its second-order variation becomes important ([Marinescu and Balcau, 2025](#)). The positive-MI comparison developed here is therefore not a replacement for a one-table

independence test. Second-order effects can also matter near that boundary.

Resampling alternatives.

A bootstrap would draw many new samples from the observed tables to estimate how much the test statistic varies. This requires repeating the calculation many times (Efron, 1979). A simple permutation test instead combines the two samples and shuffles their group labels. That shuffle is justified when the populations have the same complete distribution, $P = Q$. Our hypothesis requires only $I(P) = I(Q)$: the tables can differ while their MI values are equal. Shuffling labels could therefore test a stronger claim than this thesis asks about (Chung and Romano, 2013). More specialised permutation methods are possible for some such comparisons (Janssen, 1997; Chung and Romano, 2013), but assessing one for equal MI would be a separate study. The present comparison focuses on Normal Wald and Expanded Welch.

2.5 Previous work and this thesis’s contribution

The cited work supplies several parts of the present comparison. Moddemeyer (1989) and Brillinger (2004) give the leading sampling variance of an MI estimate; Miller (1955) and Paninski (2003) explain why finite-sample bias and sparse tables also matter. Hutcheson’s entropy comparison uses a Student reference with degrees of freedom assigned from sample sizes (Hutcheson, 1970). In a different setting, Pan and Wall (2002) show how uncertainty in a non-classical variance estimate can inform degrees of freedom. These are useful precedents, but none supplies the change in the plug-in MI variance when its observed table changes.

The question of comparing information values also has precedents. Mora and Ruiz-Castillo (2009) discusses comparisons of an MI-based segregation index. Tests based on MI between pooled observations and their group labels instead target equality of the complete distributions (Guha and Chothia, 2014; Drake and Guha, 2014). The present null is narrower: $I(P) = I(Q)$ permits $P \neq Q$. This difference in the null hypothesis determines which existing reference distributions can be used directly.

The remaining calculation is to measure how much the *estimated MI variance* changes from sample to sample. Assigning each population degrees of freedom from its sample size does not measure this change. A new observation affects a cell probability, its row and column totals, and the MI values calculated from them. This thesis derives how those changes affect the estimated variance and uses the result to choose a Student t distribution for comparing two MI values. It then tests the method on fixed population tables, showing when it brings false-positive rates

closer to the target and when it instead rejects too rarely.

Chapter 3 defines the comparison, Chapter 4 derives the adjustment, and Chapter 6 evaluates it.

Chapter 3

The Two-Sample MI Test and Wald Baseline

This chapter defines the population MI difference being tested and the statistic calculated from the observed tables. Both methods use this same statistic. Normal Wald judges it against a standard normal distribution; Chapter 4 develops a Student distribution with estimated degrees of freedom.

3.1 Problem setup and bias-corrected estimator

Let

$$N^P = (N_{ij}^P), \quad N^Q = (N_{ij}^Q) \quad (3.1)$$

be independent $r \times c$ count tables with totals n_P and n_Q . The row and column categories have the same definitions in both populations. The empirical probabilities are

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{q}_{ij} = \frac{N_{ij}^Q}{n_Q}. \quad (3.2)$$

The population parameter is the signed difference

$$\Delta = I(P) - I(Q). \quad (3.3)$$

The principal test is two-sided:

$$H_0 : \Delta = 0 \quad \text{against} \quad H_1 : \Delta \neq 0. \quad (3.4)$$

The large-sample argument uses the following assumptions.

Assumption 1 (Sampling). *Observations are independent and identically dis-*

tributed within each population, and the two population samples are independent.

Assumption 2 (Aligned fixed alphabets). *The row and column categories are fixed in advance and aligned between the two tables.*

Assumption 3 (Positive cell probabilities). *Every cell in the population table has a probability greater than zero. A sample can still contain no observations in a cell, especially when that cell is rare. Such samples remain in the study; we record how often cells have zero observed counts to show how sparse the samples are.*

Assumption 4 (Positive MI). *Both populations have positive MI: $I(P) > 0$ and $I(Q) > 0$. With the positive cell probabilities in Assumption 3, this is equivalent to requiring positive first-order variances $V(P) > 0$ and $V(Q) > 0$. It excludes exact independence, where the first-order approximation breaks down.*

Assumption 5 (Logarithm units). *All logarithms in this thesis are natural, so MI differences and standard errors are reported in nats. Changing the logarithm base scales both by the same positive constant and leaves the standardised statistic unchanged. This is a choice of units, not a condition for the test to be valid.*

Bias-corrected estimator.

For the first table, define empirical pointwise MI by

$$\hat{\ell}_P(i, j) = \log \left(\frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}} \right) \quad \text{when } \hat{p}_{ij} > 0. \quad (3.5)$$

The plug-in MI estimate is

$$\hat{I}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \hat{\ell}_P(i, j). \quad (3.6)$$

The same calculation gives $\hat{I}(Q)$.

The leading fixed-support MI bias appears in Equation (2.8) (Moddemeijer, 1989). Subtracting it from each plug-in estimate gives the estimated effect:

$$\hat{\Delta}_{\text{BC}} = \left\{ \hat{I}(P) - \frac{(r-1)(c-1)}{2n_P} \right\} - \left\{ \hat{I}(Q) - \frac{(r-1)(c-1)}{2n_Q} \right\}. \quad (3.7)$$

The two terms in braces are the bias-corrected MI estimates for the separate populations.

The correction uses the declared alphabet dimensions, not the observed nonempty dimensions. A corrected estimate can be negative in a sparse sample and is not clipped at zero, since clipping would define a different nonlinear estimator. When

$n_P = n_Q$, the two corrections cancel in the difference; otherwise, their difference removes the leading unequal-sample bias described in Section 2.2.

3.2 Shared standard error and test statistic

Section 2.3 showed that the first-order sampling variance of $\hat{I}(P)$ is $V(P)/n_P$, where $V(P)$ is the population variance of pointwise MI (Moddemeijer, 1989; Brillinger, 2004). Subtracting the fixed leading bias term in Equation (3.7) does not change this variance.

The observed table estimates $V(P)$ by

$$\hat{V}(P) = \sum_{i,j:\hat{p}_{ij}>0} \hat{p}_{ij} \{\hat{\ell}_P(i,j) - \hat{I}(P)\}^2. \quad (3.8)$$

Define $\hat{V}(Q)$ analogously.

Because the two samples are independent, their estimated variance contributions add:

$$\widehat{\text{SE}}^2 = \frac{\hat{V}(P)}{n_P} + \frac{\hat{V}(Q)}{n_Q}. \quad (3.9)$$

The common standardised statistic is

$$T = \frac{\hat{\Delta}_{\text{BC}}}{\widehat{\text{SE}}}. \quad (3.10)$$

The null hypothesis is not imposed when calculating $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$ or T : they use the observed tables. Equal population MI enters when interpreting the reference distribution, which is centred at zero under H_0 . Equations (3.7)–(3.10) are shared by Normal Wald and Expanded Welch. Their difference lies in the reference distribution, not the effect estimate or plug-in standard error.

3.3 Normal Wald reference and motivation for expansion

When both sample sizes grow while the table dimensions and positive population probabilities stay fixed, the statistic approaches a standard normal distribution under equal population MI (Wald, 1943; van der Vaart, 1998):

$$T \xrightarrow{d} N(0, 1) \quad \text{under } H_0. \quad (3.11)$$

The two-sided Normal Wald p-value is

$$p_{\text{Wald}} = 2\{1 - \Phi(|T|)\}, \quad (3.12)$$

where Φ is the standard normal distribution function.

The normal reference is a large-sample approximation. Wald estimates the standard error from the tables but applies no degrees-of-freedom adjustment.

Why consider an expanded reference?

The shared statistic supplies the estimated effect and its first-order standard error. Earlier information-theoretic and ordinary-Welch analogues set each variance component's degrees of freedom from its sample size: n_i (Hutcheson, 1970) or $n_i - 1$ (Welch, 1947). Neither assignment measures how much the plug-in MI variance changes when the observed table changes. That variance depends on joint cells, margins and pointwise MI values. Expanded Welch instead estimates each component's degrees of freedom from how the complete MI variance estimate changes with the table. The supplementary analysis also compares these sample-size assignments with Expanded Welch.

To choose the degrees of freedom, we need the average value of each estimated MI variance and how much that estimate varies across samples (Satterthwaite, 1946). For P , to first order,

$$E\{\hat{V}(P)\} \approx V(P). \quad (3.13)$$

Chapter 4 derives $\text{Var}\{\hat{V}(P)\}$ by tracking how the calculated MI variance changes when one observation changes the joint table.

Chapter 4

The Expanded Welch–Satterthwaite Test for MI

Expanded Welch keeps the estimated MI difference, standard error and test statistic from Chapter 3. It changes the distribution used to judge that statistic from normal to Student. To choose the Student degrees of freedom, this chapter calculates how much the estimated MI variance changes between samples.

4.1 Choosing Degrees of Freedom for Expanded Welch

For the two MI estimates, the Welch–Satterthwaite equation combines their estimated variance contributions as in the original moment-matching framework (Satterthwaite, 1946; Welch, 1947):

$$\nu = \frac{\left\{ \hat{V}(P)/n_P + \hat{V}(Q)/n_Q \right\}^2}{\left\{ \hat{V}(P)/n_P \right\}^2 / \nu_P + \left\{ \hat{V}(Q)/n_Q \right\}^2 / \nu_Q}. \quad (4.1)$$

The numerator contains the two terms in the squared standard error. The remaining quantities are the component degrees of freedom ν_P and ν_Q , which should measure how reliably $\hat{V}(P)$ and $\hat{V}(Q)$ estimate their population counterparts.

The derivation has four steps. First, we relate the degrees of freedom to the average and across-sample variation of $\hat{V}(P)$. Second, we calculate how $V(P)$ changes when a small amount of probability moves into one cell. Third, that change tells us how variable $\hat{V}(P)$ is. Finally, we combine the degrees of freedom from P and Q using Equation (4.1).

Moment matching for one component.

Satterthwaite represents a positive variance estimate by a scaled chi-squared variable. For the first population, the working approximation represents $\widehat{V}(P)$ by

$$E\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}. \quad (4.2)$$

The approximation matches moments; it does not assert that the finite-sample distribution of $\widehat{V}(P)$ is exactly chi-squared.

For a chi-squared variable with ν_P degrees of freedom,

$$E(\chi_{\nu_P}^2) = \nu_P, \quad \text{Var}(\chi_{\nu_P}^2) = 2\nu_P. \quad (4.3)$$

The scaling in Equation (4.2) therefore matches the mean automatically:

$$E\left[E\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}\right] = E\{\widehat{V}(P)\}. \quad (4.4)$$

Its variance is

$$\text{Var}\left[E\{\widehat{V}(P)\} \frac{\chi_{\nu_P}^2}{\nu_P}\right] = \frac{[E\{\widehat{V}(P)\}]^2}{\nu_P^2} \text{Var}(\chi_{\nu_P}^2) \quad (4.5)$$

$$= \frac{2[E\{\widehat{V}(P)\}]^2}{\nu_P}. \quad (4.6)$$

Matching this quantity to $\text{Var}\{\widehat{V}(P)\}$ gives

$$\boxed{\nu_P = \frac{2[E\{\widehat{V}(P)\}]^2}{\text{Var}\{\widehat{V}(P)\}}}. \quad (4.7)$$

To first order, $E\{\widehat{V}(P)\} \approx V(P)$. The required calculation is therefore the sampling variance of $\widehat{V}(P)$.

4.2 How the MI Variance Changes with the Table

To find how much the estimated MI variance changes between samples, we first ask how the population variance responds when a small amount of probability is moved into one cell (Hampel, 1974; van der Vaart, 1998). Fix a cell (x, y) and define the perturbed distribution

$$P_\varepsilon = (1 - \varepsilon)P + \varepsilon\delta_{(x,y)}, \quad (4.8)$$

where $\delta_{(x,y)}$ places probability one on cell (x, y) . At $\varepsilon = 0$, the distribution is P . Increasing ε moves a small amount of probability toward the selected cell while preserving total probability one.

For an arbitrary cell (i, j) ,

$$p_{ij}(\varepsilon) = (1 - \varepsilon)p_{ij} + \varepsilon \mathbf{1}\{i = x, j = y\}. \quad (4.9)$$

The joint and marginal probability derivatives at the original distribution are

$$\left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x, j = y\} - p_{ij}, \quad (4.10)$$

$$\left. \frac{dp_{i+}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{i = x\} - p_{i+}, \quad (4.11)$$

$$\left. \frac{dp_{+j}(\varepsilon)}{d\varepsilon} \right|_0 = \mathbf{1}\{j = y\} - p_{+j}. \quad (4.12)$$

The derivatives of the joint probabilities sum to zero because

$$\sum_{i,j} \{\mathbf{1}(i = x, j = y) - p_{ij}\} = 1 - 1 = 0. \quad (4.13)$$

Effect on pointwise MI and MI.

Along the perturbation path, pointwise MI is

$$\ell_{P_\varepsilon}(i, j) = \log p_{ij}(\varepsilon) - \log p_{i+}(\varepsilon) - \log p_{+j}(\varepsilon). \quad (4.14)$$

Differentiating the three logarithms and substituting the probability derivatives gives

$$\left. \frac{d}{d\varepsilon} \ell_{P_\varepsilon}(i, j) \right|_0 = \frac{\mathbf{1}\{i = x, j = y\}}{p_{ij}} - \frac{\mathbf{1}\{i = x\}}{p_{i+}} - \frac{\mathbf{1}\{j = y\}}{p_{+j}} + 1. \quad (4.15)$$

MI along the path is

$$I(P_\varepsilon) = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j). \quad (4.16)$$

The product rule separates its derivative into the change in probability weights and the change in pointwise-MI values:

$$\begin{aligned} \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 &= \sum_{i,j} \{ \mathbf{1}(i=x, j=y) - p_{ij} \} \ell_P(i, j) \\ &\quad + \sum_{i,j} p_{ij} \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.17)$$

The first sum is $\ell_P(x, y) - I(P)$. In the second sum, the selected-cell term contributes $p_{xy}/p_{xy} = 1$. The row term contributes $-\sum_j p_{xj}/p_{x+} = -1$; the column term likewise contributes -1 ; and the constant contributes $\sum_{i,j} p_{ij} = 1$. These four contributions cancel. Hence

$$\boxed{\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P)}. \quad (4.18)$$

This result recovers the centred pointwise-MI influence function used in Chapter 3.

Effect on the MI variance.

The target is the first-order change in

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\}. \quad (4.19)$$

Write this variance in second-moment form:

$$V(P) = \text{E}_P\{\ell_P(X, Y)^2\} - I(P)^2. \quad (4.20)$$

Define the observation-level variance sensitivity by

$$g_P(x, y) = \left. \frac{dV(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.21)$$

Differentiating Equation (4.20) gives

$$g_P(x, y) = \left. \frac{d}{d\varepsilon} \text{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} \right|_0 - 2I(P) \left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0. \quad (4.22)$$

It remains to differentiate

$$\text{E}_{P_\varepsilon}\{\ell_{P_\varepsilon}(X, Y)^2\} = \sum_{i,j} p_{ij}(\varepsilon) \ell_{P_\varepsilon}(i, j)^2. \quad (4.23)$$

The product rule gives

$$\begin{aligned} \left. \frac{d}{d\varepsilon} E_{P_\varepsilon} \{ \ell_{P_\varepsilon}(X, Y)^2 \} \right|_0 &= \sum_{i,j} \left. \frac{dp_{ij}(\varepsilon)}{d\varepsilon} \right|_0 \ell_P(i, j)^2 \\ &\quad + 2 \sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0. \end{aligned} \quad (4.24)$$

The first sum is

$$\ell_P(x, y)^2 - E_P \{ \ell_P(X, Y)^2 \}. \quad (4.25)$$

For the second sum, define the probability-weighted row and column means

$$E_P \{ \ell_P(X, Y) \mid X = x \} = \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}}, \quad (4.26)$$

$$E_P \{ \ell_P(X, Y) \mid Y = y \} = \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}}. \quad (4.27)$$

Substitution of Equation (4.15) gives

$$\begin{aligned} &\sum_{i,j} p_{ij} \ell_P(i, j) \left. \frac{d\ell_{P_\varepsilon}(i, j)}{d\varepsilon} \right|_0 \\ &= \ell_P(x, y) - \frac{\sum_j p_{xj} \ell_P(x, j)}{p_{x+}} - \frac{\sum_i p_{iy} \ell_P(i, y)}{p_{+y}} + I(P) \\ &= \ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P). \end{aligned} \quad (4.28)$$

Combining Equations (4.18), (4.22), (4.25), and (4.28) yields

$$\begin{aligned} g_P(x, y) &= \{ \ell_P(x, y) - I(P) \}^2 - V(P) \\ &\quad + 2 \left[\ell_P(x, y) - E_P \{ \ell_P(X, Y) \mid X = x \} \right. \\ &\quad \left. - E_P \{ \ell_P(X, Y) \mid Y = y \} + I(P) \right]. \end{aligned} \quad (4.29)$$

The first line is the change if the pointwise-MI values were held fixed. The bracket accounts for changes in those values, including the effects of the selected joint probability and both margins.

4.3 Component degrees of freedom and final test

Population component degrees of freedom.

Define the population variability of the observation-level sensitivities by

$$\tau^2(P) = \text{Var}_P\{g_P(X, Y)\}. \quad (4.30)$$

Averaging the one-cell perturbation directions gives $\sum_{x,y} p_{xy}\{\delta_{(x,y)} - P\} = 0$. The derivative is linear in its direction, so $E_P\{g_P(X, Y)\} = 0$; Appendix A shows the steps. Explicit centring protects the numerical calculation if roundoff produces a small nonzero empirical mean.

Assumption 6 (Regular variance sensitivity). *For the first-order Expanded Welch construction, $\tau^2(P) > 0$ and $\tau^2(Q) > 0$.*

The empirical distribution is

$$\hat{P} - P = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\delta_{Z_a^{(P)}} - P\}. \quad (4.31)$$

Taylor's first-order term is linear in this difference (van der Vaart, 1998, Chapter 20). By Equation (4.21), each summand has sensitivity $g_P(Z_a^{(P)})$. Therefore,

$$\hat{V}(P) - V(P) \approx \frac{1}{n_P} \sum_{a=1}^{n_P} g_P(Z_a^{(P)}). \quad (4.32)$$

The observations are independent, so

$$\text{Var}\{\hat{V}(P)\} \approx \frac{1}{n_P^2} \sum_{a=1}^{n_P} \text{Var}_P\{g_P(Z_a^{(P)})\} \quad (4.33)$$

$$= \frac{\tau^2(P)}{n_P}. \quad (4.34)$$

Substitution into Equation (4.7) gives the population component degree of freedom

$$\boxed{\nu_P \approx \frac{2n_P V(P)^2}{\tau^2(P)}}. \quad (4.35)$$

Equation (4.35) depends on relative, not absolute, uncertainty. Since

$$\frac{\text{Var}\{\hat{V}(P)\}}{V(P)^2} \approx \frac{\tau^2(P)}{n_P V(P)^2}, \quad (4.36)$$

the component degrees of freedom are approximately twice the reciprocal of this relative variance. A stable estimate relative to the size of $V(P)$ has large degrees of freedom; a relatively unstable estimate has smaller degrees of freedom and produces a heavier-tailed final reference. This first-order component formula assumes $\tau^2(P) > 0$.

Positive $V(P)$ alone does not guarantee that condition. A small change in the table can have no first-order effect on $V(P)$, even when MI and $V(P)$ are positive. In that case, a higher-order analysis of $\hat{V}(P)$ would be needed to describe its finite-sample distribution.

Table-based component degrees of freedom.

The observed table replaces every population quantity in Equation (4.29) by its empirical counterpart. The sums below are over occupied cells, because pointwise MI is undefined at an empirical zero. For each occupied cell,

$$\begin{aligned} \hat{g}_P(i, j) = & \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \\ & + 2 \left[\hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right]. \end{aligned} \quad (4.37)$$

In all weighted sums, an empty cell contributes zero; no logarithm or sensitivity needs to be evaluated there. The empirical variance is centred explicitly:

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2. \quad (4.38)$$

The estimated component degree of freedom is

$$\boxed{\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}}. \quad (4.39)$$

Repeating the same calculation for the second table gives $\hat{\nu}_Q$.

Combining the two components.

Substituting the two variance contributions and component degrees of freedom into Equation (4.1) gives

$$\boxed{\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\{\hat{V}(P)/n_P\}^2 / \hat{\nu}_P + \{\hat{V}(Q)/n_Q\}^2 / \hat{\nu}_Q}}. \quad (4.40)$$

The expanded two-sided p-value is

$$p_{\text{expanded}} = 2 \left\{ 1 - F_{t_{\hat{\nu}_{\text{expanded}}}}(|T|) \right\}, \quad (4.41)$$

where F_{t_ν} denotes the cumulative distribution function of a Student t variable with ν degrees of freedom.

The expanded calculation does not change $\hat{\Delta}_{\text{BC}}$, $\widehat{\text{SE}}$, or T . It changes only the

reference distribution used to evaluate that statistic.

For a fixed finite T , the Student reference gives a two-sided p-value at least as large as the Normal Wald reference. Under the implemented validity rules, every valid Expanded Welch result is also Wald-valid. Thus every Expanded Welch rejection is also a Wald rejection, including when invalid outputs count as non-rejections in the reported rates. The correction cannot raise rejection where Wald is already too conservative. Appendix A proves this ordering and examines a zero-component case in the combined degrees-of-freedom formula.

4.4 Worked example and implementation

Consider the two observed 3×3 tables

$$N^P = \begin{pmatrix} 35 & 10 & 5 \\ 10 & 20 & 5 \\ 5 & 5 & 10 \end{pmatrix}, \quad N^Q = \begin{pmatrix} 30 & 5 & 5 \\ 5 & 30 & 5 \\ 5 & 5 & 10 \end{pmatrix}. \quad (4.42)$$

Their totals are $n_P = 105$ and $n_Q = 100$. The plug-in MI estimates are 0.13693 and 0.25848 nats. Since $(3 - 1)(3 - 1) = 4$, the corrected difference is

$$\hat{\Delta}_{\text{BC}} = \left(0.13693 - \frac{4}{2(105)}\right) - \left(0.25848 - \frac{4}{2(100)}\right) = -0.12059. \quad (4.43)$$

The empirical influence variances are $\hat{V}(P) = 0.27189$ and $\hat{V}(Q) = 0.43371$. Hence

$$\widehat{\text{SE}} = \left(\frac{0.27189}{105} + \frac{0.43371}{100}\right)^{1/2} = 0.08323, \quad T = -1.44898. \quad (4.44)$$

Normal Wald gives $p = 0.14734$.

For Expanded Welch, the table-based sensitivity calculation first gives $\hat{\tau}^2(P) = 0.93212$ and $\hat{\tau}^2(Q) = 0.82035$. Substitution into Equation (4.39) gives component degrees of freedom

$$\hat{\nu}_P = 16.65, \quad \hat{\nu}_Q = 45.86. \quad (4.45)$$

Equation (4.40) combines them into $\hat{\nu}_{\text{expanded}} = 59.03$, giving $p = 0.15264$. Both methods fail to reject at the 5% level for these tables. The example shows that Expanded Welch changes the reference, not the observed statistic.

Operational algorithm and complexity.

Table 4.1 summarises one comparison.

Table 4.1: Operational Expanded Welch–Satterthwaite algorithm.

Step	Calculation
1	Validate the two aligned count tables and their declared alphabet dimensions; independence of samples is a design assumption.
2	Compute empirical probabilities, pointwise MI, plug-in MI, and the leading bias-corrected difference.
3	Compute $\hat{V}(P)$, $\hat{V}(Q)$, the shared standard error, and T .
4	Compute the row and column pointwise-MI means needed in $\hat{g}_P$ and $\hat{g}_Q$.
5	Compute $\hat{\tau}^2(P)$, $\hat{\tau}^2(Q)$, and the two component degrees of freedom.
6	Combine the components using Equation (4.40).
7	Return the Student p-value, effective degrees of freedom, and validity diagnostics.

Each empirical probability, pointwise-MI value, row mean, column mean, and variance contribution is calculated by a fixed number of table scans. The time complexity is therefore $O(rc)$, and the working memory is also $O(rc)$. The method is deterministic once the two count tables are given.

4.5 Behaviour and limitations

Near independence.

The adjustment has a local explanation. Let $P_t = P_0 + tH$, where $P_0 = (q_{ij})$ is independent with positive cells $q_{ij} = a_i b_j$, and H has zero row and column sums. Thus $t = 0$ is the expansion point and the margins stay fixed. For $f(t) = I(P_t)$, the second-order Taylor expansion has $f(0) = 0$, $f'(0) = \sum_{i,j} H_{ij} = 0$, and $f''(0) = \sum_{i,j} H_{ij}^2 / q_{ij}$.

$$I(P_t) = \underbrace{0}_{\text{population value}} + \underbrace{0t}_{\text{first order}} + \underbrace{\frac{t^2}{2} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}}}_{\text{second order}} + O(t^3), \quad (4.46)$$

$$\ell_{P_t}(i, j) = \log \left(1 + \frac{tH_{ij}}{q_{ij}} \right) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \quad (4.47)$$

$$V(P_t) = t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.48)$$

Appendix E gives the second-order MI result when the margins can also change. The row and column conditional means of pointwise MI have no first-order term because the corresponding sums of H are zero. Equation (4.29) then gives $g_{P_t}(i, j) =$

$2tH_{ij}/q_{ij} + O(t^2)$, and hence

$$\tau^2(P_t) = 4t^2 \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O(t^3). \quad (4.49)$$

For a distribution close to independence on this path, these expansions give

$$V(P) \approx 2I(P), \quad \tau^2(P) \approx 4V(P). \quad (4.50)$$

Substitution into the component formula then gives

$$\nu_P = \frac{2n_P V(P)^2}{\tau^2(P)} \approx n_P I(P). \quad (4.51)$$

Thus low MI can produce a small effective degree of freedom and a much heavier-tailed reference. The approximation is local: it assumes fixed table dimension, positive probabilities and sufficiently regular cell counts. Very near independence, quadratic sampling variation can still be important, so a first-order Student adjustment need not improve finite-sample calibration.

Theoretical status and validity conditions.

Equations (4.18) and (4.29) describe how MI and its estimated variance change when the underlying table changes slightly. For tables with positive cell probabilities, these changes give their first-order large-sample approximations.

The scaled chi-squared model in Equation (4.2) and the Student reference are approximations, not exact finite-sample distributions. In particular, $\hat{I}(P)$ and $\hat{V}(P)$ come from the same sample and can vary together. Adjusting the reference for uncertainty in $\hat{V}(P)$ does not account fully for this dependence or make the statistic exactly Student distributed. Appendix A gives the first-order covariance. As sample sizes grow under regular conditions, the estimated degrees of freedom increase and the Student reference approaches the normal reference used by Wald.

Both tests need a finite corrected MI difference and a positive, finite estimated standard error. Expanded Welch also needs positive, finite degrees of freedom for each population and for their combination. If a requirement fails, the calculation is recorded as invalid, not as evidence for the null. Appendix F gives the exact software rules. Chapters 5 and 6 report how often valid results are obtained and whether the correction improves calibration.

Chapter 5

Experimental Design

The purpose of the experiment is to compare the calibration, detection and validity of two tests across controlled populations. Each design point fixes two population distributions in advance. Repeated independent samples from those populations show how often each procedure returns a valid result and how often it rejects. Because the population MI values are known, a rejection at equal MI is a false positive; a rejection at unequal MI is detection.

5.1 A Concrete Pair of Population Tables

Consider two 2×2 populations whose row and column margins are $(0.7, 0.3)$ for P and $(0.8, 0.2)$ for Q . The construction produces, to the displayed precision,

$$P \simeq \begin{pmatrix} 0.53273 & 0.16727 \\ 0.16727 & 0.13273 \end{pmatrix}, \quad Q \simeq \begin{pmatrix} 0.68175 & 0.11825 \\ 0.11825 & 0.08175 \end{pmatrix}. \quad (5.1)$$

These are *probability* tables, not observed counts. Their target MI values are 0.02 and 0.03 nats, so the true magnitude of the difference is 0.01 nats. Sampling new count tables changes the estimated MI values, but does not move this fixed population pair to a new horizontal-axis location. Calculations use the stored unrounded probabilities; the rounded display in Equation (5.1) does not determine the samples.

5.2 Constructing fixed joint distributions

For one population, let $a = (a_1, \dots, a_r)$ and $b = (b_1, \dots, b_c)$ be prescribed row and column probabilities. The independence table has cell probabilities $a_i b_j$. Association

is introduced by

$$R(t) = ab^T + tH, \quad H\mathbf{1}_c = 0, \quad \mathbf{1}_r^T H = 0. \quad (5.2)$$

The zero row and column sums mean that changing t leaves the margins fixed. For the main construction, H is zero except in the first two rows and columns, where it is

$$H_{1:2,1:2} = \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}. \quad (5.3)$$

Thus t is added to the two diagonal cells and removed from the other two cells. The changes cancel within each affected row and column. The same matrix specifies a 2×2 association and a localised change in a larger table; only its embedding differs. This follows the minimal four-violation sparse alternative used by [Berrett and Samworth \(2021, Section 3\(b\)\)](#). Here it is adapted to compare two fixed populations: their margins are prescribed separately and the perturbation strength is chosen to attain a specified MI and is not used directly as the horizontal-axis effect.

For the pair in Equation (5.1), the starting independence tables are

$$P_0 = \begin{pmatrix} 0.49 & 0.21 \\ 0.21 & 0.09 \end{pmatrix}, \quad Q_0 = \begin{pmatrix} 0.64 & 0.16 \\ 0.16 & 0.04 \end{pmatrix}. \quad (5.4)$$

Adding the block in Equation (5.3) with strengths $t_P \simeq 0.04273$ and $t_Q \simeq 0.04175$, respectively, gives the displayed P and Q . The strengths are chosen to reach their separate MI targets; the 0.01-nat difference is not a cell-probability increment.

The largest nonnegative strength before a cell with $H_{ij} < 0$ reaches zero is

$$t_{\max} = \min_{H_{ij} < 0} \frac{a_i b_j}{-H_{ij}}. \quad (5.5)$$

The construction solves for $0 \leq t < t_{\max}$ such that $I\{R(t)\}$ equals the requested target. Targets at or beyond the endpoint are not admitted. This endpoint is specific to the chosen path, not the largest possible MI over every table with those margins. Both constructed tables are checked for strictly positive cells, correct margins and target MI before any sampling. Appendix B gives the numerical details.

For a dimension of size k , the skewed margin used in the design is

$$m_k(\rho) = \left(\rho, \frac{1-\rho}{k-1}, \dots, \frac{1-\rho}{k-1} \right). \quad (5.6)$$

The three principal profiles are specified separately for the two populations in

Table 5.1. For a rectangular $r \times c$ table, the formula is applied with $k = r$ to row margins and $k = c$ to column margins. At a zero MI difference, the same-margin profiles produce $P = Q$ under this construction; the different-skew profile has $I(P) = I(Q)$ but $P \neq Q$.

Table 5.1: Row and column margins used for each population in the principal design. “Uniform” means the vector with every entry equal to $1/k$.

Profile	P : rows and columns	Q : rows and columns
Uniform	uniform	uniform
Same skew	$m_k(0.8)$	$m_k(0.8)$
Different skew	$m_k(0.7)$	$m_k(0.8)$

In the usual effect direction, $I(P) = I_0$ and $I(Q) = I_0 + |\Delta|$. The paired solve chooses separate strengths for P and Q ; it does not add random noise to their true MI values. The reverse-direction imbalance experiment assigns the higher target to P . The statistical convention remains $\Delta = I(P) - I(Q)$, which is negative in the usual direction. Figures use the nonnegative magnitude $|\Delta|$.

Two focused checks ask whether results depend on where the association sits in the table. The “rare block” moves the four changed cells from the upper-left corner to rows and columns with smaller probabilities in skewed populations. The “spread” pattern changes cells throughout the table while keeping the row and column totals fixed. The log-linear comparison also spreads association through the table, using a different rule. In each case, the population is built to have the specified margins and target MI. Tables can therefore agree on margins and MI while assigning different probabilities to individual cells. Appendix B gives the three construction formulas.

5.3 Sampling, tests and measured outcomes

For each unique configuration, 20,000 independent pairs of multinomial count tables are drawn from its fixed P and Q . Normal Wald and Expanded Welch are applied to the *same* count-table pair. This pairing makes method differences more precisely measurable than unrelated simulations, though it does not remove Monte Carlo uncertainty. The two-sided nominal threshold is $\alpha = 0.05$ throughout the statistical study.

For each method, the primary plotted rejection rate is

$$\frac{\#\{\text{sample pairs giving a valid p-value no larger than } 0.05\}}{20,000}. \quad (5.7)$$

At $|\Delta| = 0$ this estimates the false-positive rate. At $|\Delta| > 0$ it estimates the probability of detecting the specified difference at the fixed nominal threshold, often called power. A rate near 0.05 is desirable under the null; a smaller rate is not automatically better. Under an alternative, higher detection is useful only alongside acceptable null calibration and a usable test result.

An invalid calculation contributes no rejection to Equation (5.7), but it does not count as evidence for the null. The valid-result rate shows how often each test returns a usable p-value. The reported 95% intervals show simulation uncertainty at each fixed population pair. Intervals for the difference between methods account for applying both tests to the same sampled tables. Appendix C gives the interval calculations and rates restricted to valid results.

The “minimum expected count” diagnostic is $\min_{ij} n_P p_{ij}$ for P , or $\min_{ij} n_Q q_{ij}$ for Q . It is not the expectation of the smallest observed count. No lower bound on that diagnostic was used to admit statistical configurations. Empty observed cells and rows remain possible. In the plots a hollow marker warns that fewer than 90% of outputs of that method were valid; 90% is a display convention, not a validated safety threshold.

5.4 Experiment families and computation

Table 5.2 records the question answered by each family. The common population construction above applies unless the named factor changes. Exact parameter sets are in Appendix B, and the machine-readable protocol is identified in Appendix F. The principal grid crosses square shapes 2×2 , 3×3 , 5×5 , 8×8 with the three margin profiles in Table 5.1. It fixes $I(P) = 0.02$ nats and uses $|\Delta| \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02\}$ nats and equal sample sizes $n_P = n_Q \in \{2, 5, 10, 20, 50, 100, 250, 500, 1000\}$. For every feasible combination, the constructor fixes P and Q before the 20,000 table pairs are sampled and both tests are applied to each pair. The other families change one or more stated factors from this principal grid; Appendix B lists their exact values. There are 534 distinct population pairs (P, Q) , each specifying two fixed joint probability tables. Reusing a pair at different sample sizes creates distinct sampling configurations. There are 3,111 unique statistical configurations but 4,001 display points: some configurations are deliberately shown in more than one comparison. Each unique configuration is sampled once, giving 62,220,000 table pairs in the final statistical run. The within-family counts in Table 5.2 do not sum to 3,111 because the same configuration can occur in several families. Display reuse is not an independent replication. The supplementary results report contains the full set of plotted points; Appendix C identifies that report and the per-configuration data

files.

Table 5.2: The fixed experiment families. All statistical points use 20,000 paired replicates and the 5% threshold.

Family	Displays	Unique	Factor examined
Main	648	648	Square table size, three margin profiles and equal sample size.
Baseline	648	648	Starting positive MI of 0.0001, 0.001 or 0.02 nats.
Changed cells	144	144	First, rare or spread association patterns.
Imbalance	1,152	1,056	Sample allocation and which population has higher MI.
Larger effects	162	162	MI differences up to 0.2 nats where feasible.
Rectangular	108	108	2×3 and 3×5 tables.
Construction	432	432	Additive versus ordinal log-linear population tables.
Extreme skew	315	315	Dominant margins up to 0.9995, including $n = 1$ stress.
Rare-cell stress	140	140	First- versus rare-block changes, including $n = 1$ stress.
Independence	144	144	Exact $I(P) = 0$ boundary.
Convergence	108	108	Fixed equal-MI populations with sample sizes up to 50,000.
Runtime	60	60	Full-call timing regimes, separate from simulation.

The runtime experiment measures a full method call after count tables are already available. It varies table shape, profile, sample size and null or alternative status. Each regime uses 200 saved inputs, 20 warm-up calls and five timed passes. Thus it measures method computation, not the cost of constructing populations or generating random tables. Median call times and interquartile ranges are reported in Chapter 6.

Chapter 6

Experimental Results and Discussion

The figures plot how often each test rejects across 20,000 sampled table pairs. A point at a true MI difference of zero measures false positives; points to its right measure detection when the population MIs differ. The horizontal coordinate is fixed by the population construction, not estimated from the random samples. Blue solid circles show Normal Wald and magenta dashed squares show Expanded Welch. The dotted line marks the target 0.05 null rejection rate. Every plotted rate divides valid rejections by all 20,000 replicates; a hollow marker warns that fewer than 90% of calculations were valid.

The comparisons proceed from null calibration to detection, sparse-sample validity, possible causes of error and larger-sample behaviour. Additional matched figures are in [Appendix D](#); [Appendix C](#) identifies the per-configuration results.

6.1 Calibration across table size and margins

The first comparison keeps both populations uniform in a 2×2 table. [Figure 6.1](#) shows three equal sample sizes and asks whether the lower Expanded Welch rejection rate is closer to the target 0.05 at each size.

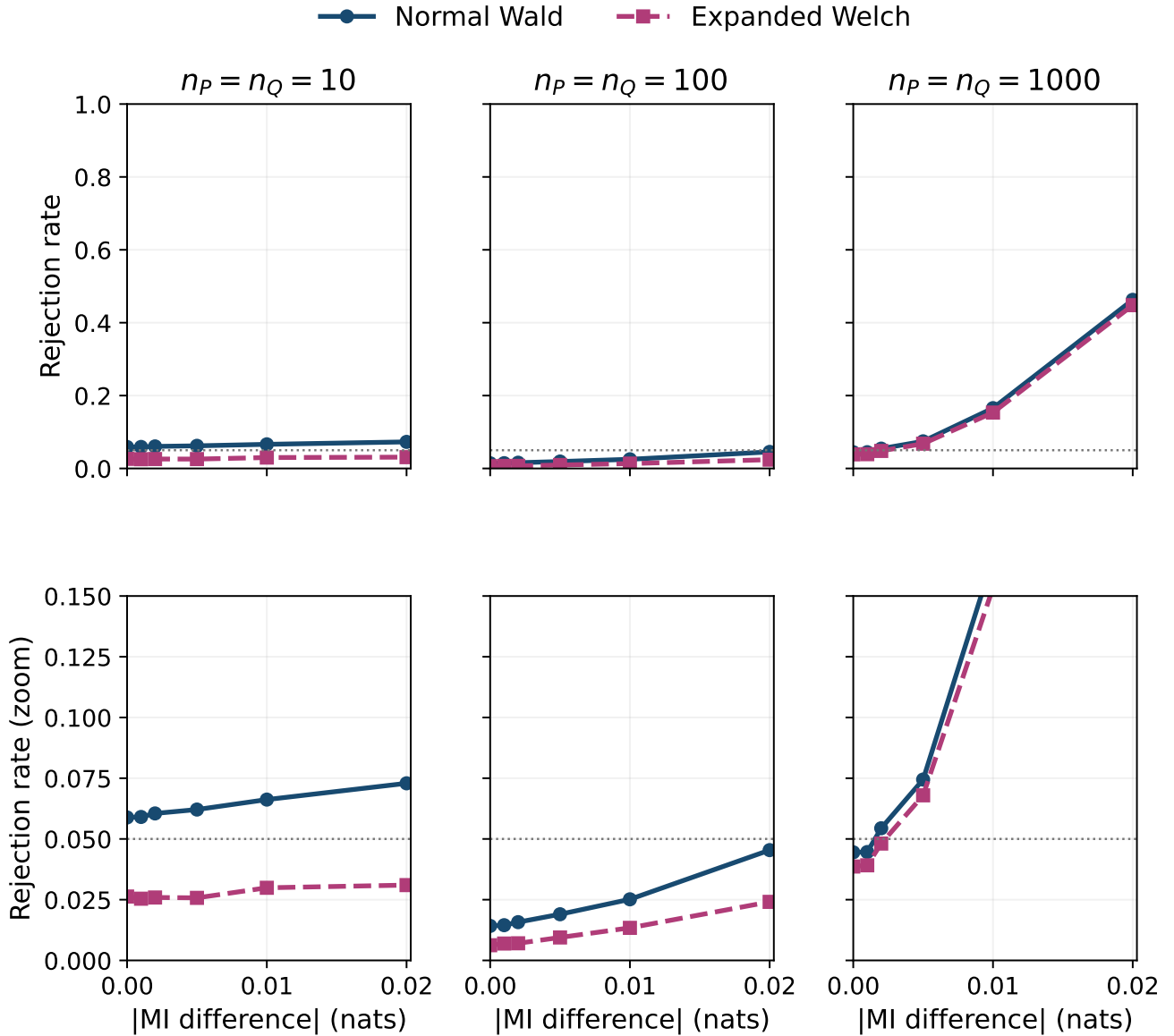


Figure 6.1: 2×2 tables, both P and Q have uniform row and column margins $(1/2, 1/2)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. At zero difference, both methods reject below 0.05 at $n = 100$ and 1,000; Expanded Welch is lower still. At $n = 10$, Wald is slightly above 0.05 and Expanded Welch below it. Validity is high in all three panels. Rejection grows most clearly with MI difference at $n = 1,000$, where the two curves are close. At the smaller sizes, low rejection should not be mistaken for better detection.

Figure 6.2 asks whether the same comparison changes in a larger, still-uniform 8×8 table.

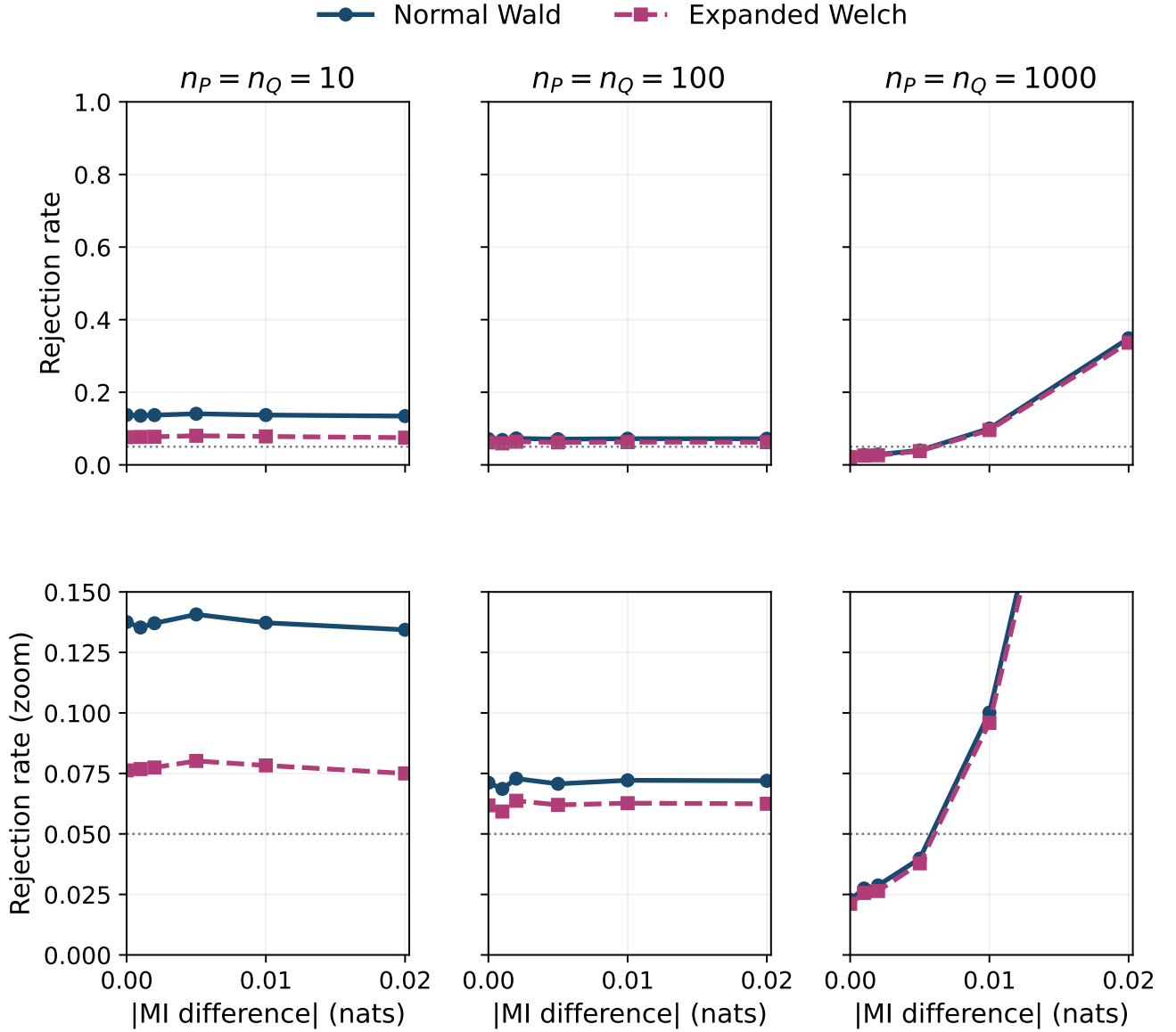


Figure 6.2: 8×8 tables, both P and Q have uniform row and column margins $(1/8, \dots, 1/8)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. At $n = 10$ and 100 , the zero-difference rates exceed 0.05 for both methods; Expanded Welch reduces the excess but does not always reach the target. These plotted calculations are valid, so the difference is not caused by missing outputs. At $n = 100$, rejection changes little over the displayed alternatives; at $n = 1,000$, it grows strongly. Even with uniform margins, table size and sample size alter the trade-off. The next comparison allows the two populations' margins to differ.

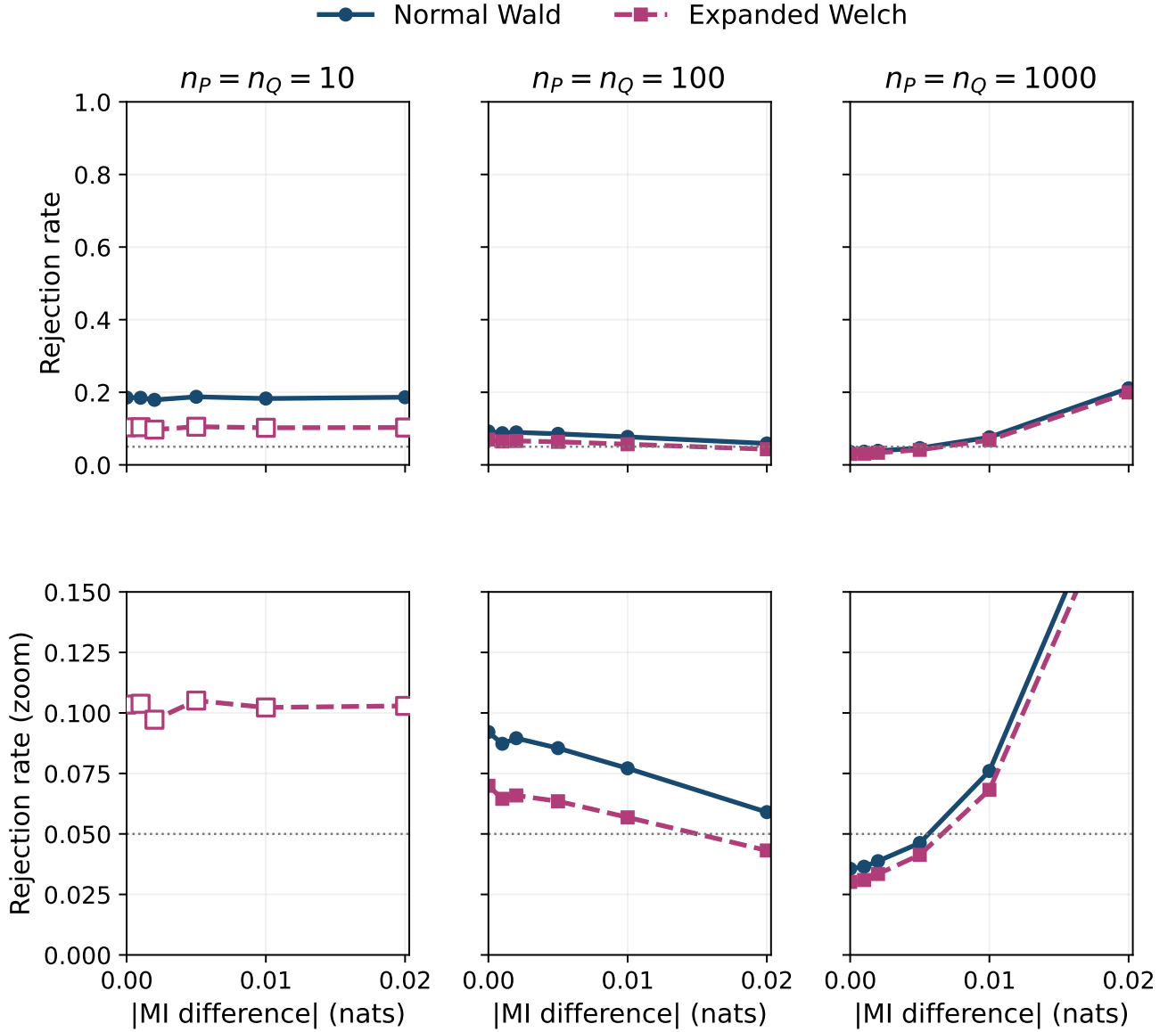


Figure 6.3: 8×8 tables, P has row and column margins $m_8(0.7) = (0.7, 0.3/7, \dots, 0.3/7)$, while Q has $m_8(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. At zero difference, both methods are too liberal at $n = 10$ and 100, although Expanded Welch rejects less. Its $n = 10$ markers are hollow, so validity contributes to that reduction. At $n = 100$, rejection falls rather than grows over the displayed alternative range; at $n = 1,000$, detection rises but stays lower for Expanded Welch. This is the clearest reminder not to infer power from the null rate alone.

Across the 24 main null regimes with $n = 10$ or 20, Expanded Welch is numerically closer to 0.05 in 15, whereas Wald is closer in 9. At $n = 500$ or 1,000, Wald is closer in all 24 main null regimes. The $n = 5$ regimes need separate treatment below because Expanded Welch never rejects there. Appendix D shows the remaining shape–margin plots.

Table 6.1 makes the two directions of the calibration trade-off concrete in null settings where both methods return a p-value on every sample.

Table 6.1: Two fully valid main-grid null regimes. Brackets beside each rate are pointwise Wilson 95% intervals; the final bracket is the approximate paired 95% interval for Wald minus Expanded Welch on the same samples. Source configurations are `config_313a5e5c05280f9e` and `config_700945d218841f2e`, respectively.

Regime	Wald	Expanded	Wald minus Expanded
	Rate [95% interval]	Rate [95% interval]	Rate [95% interval]
8×8 , uniform, $n = 20$	0.0687 [0.0653, 0.0723]	0.0570 [0.0538, 0.0602]	0.0118 [0.0103, 0.0132]
2×2 , uniform, $n = 1000$	0.0445 [0.0417, 0.0474]	0.0387 [0.0361, 0.0414]	0.0059 [0.0048, 0.0069]

In the 8×8 row, Expanded Welch moves Wald’s above-target rate closer to 0.05. In the 2×2 row, it moves an already-low rate farther below the target. Invalid outputs cannot explain either difference. The reason for this one-sided trade-off is built into the methods: on the same valid table pair, Expanded Welch uses the same statistic as Wald but a Student reference with a no-smaller two-sided p-value. It can lower rejection, not raise it. This makes detection under unequal MI the next question.

6.2 Detection as the true MI difference grows

The main grid shows only differences up to 0.02 nats. Figure 6.4 extends a uniform 2×2 comparison to 0.2 nats to check whether the low rejection over small effects persists as the true MI difference grows.

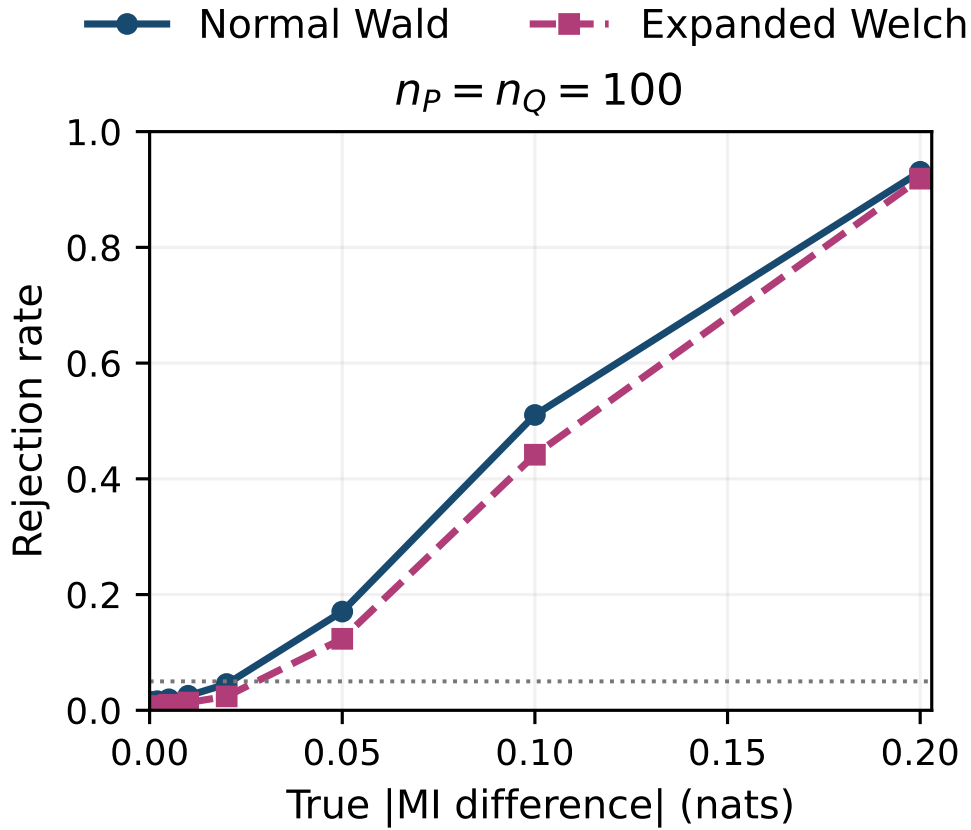


Figure 6.4: Wider MI differences in a uniform 2×2 population pair. The single panel has P and Q row and column margins $(0.5, 0.5)$, first-cell changes, $n_P = n_Q = 100$, $I(P) = 0.02$, and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2$ nats; the rejection axis is 0–1. Each point has 20,000 replicates.

Interpretation. At zero difference, both methods are below the 0.05 line; Expanded Welch is lower, and validity is high. Rejection then rises markedly as the true MI difference reaches the larger targets: by 0.2 nats both methods reject more than 90% of samples. Expanded Welch remains below Wald at intermediate differences, but the gap narrows when detection is already high. This wider axis shows that the low rejection over the main grid’s small differences is not a permanent ceiling on detection.

The wider effect range establishes that the low rejection in the main grid is not a permanent ceiling on detection. The starting association may also matter. Figure 6.5 repeats a 3×3 different-skew comparison with three starting MI values while keeping the other displayed settings fixed.

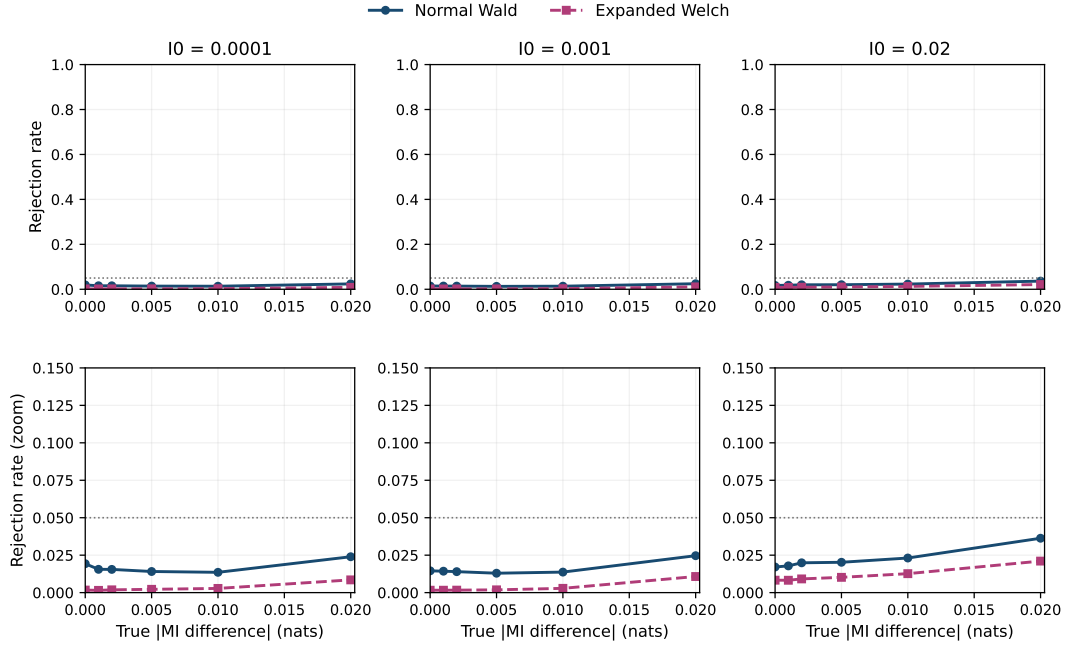


Figure 6.5: Baseline-MI comparison for 3×3 tables. Columns have $I(P) = 0.0001, 0.001, 0.02$ nats, respectively; $I(Q) = I(P) + |\Delta|$. P has row and column margins $(0.7, 0.15, 0.15)$ and Q has $(0.8, 0.1, 0.1)$. Both use the first-cell change block, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1 rejection scale; bottom: 0–0.15 zoom. Each point has 20,000 replicates; hollow markers flag validity below 90%.

Interpretation. For $I(P) = 0.0001, 0.001, 0.02$, the zero-difference Wald rates are 0.0193, 0.0146 and 0.0171; Expanded Welch gives 0.0017, 0.0016 and 0.0083. All these null calculations are valid. Rejection rises only modestly across the displayed alternatives, and both methods remain mostly below 0.05. Starting closer to independence does not make the larger Student cutoff helpful here: it lowers an already-conservative null rate and further reduces detection.

The local calculation in Chapter 4 explains why its degrees of freedom can become small near independence: the Student cutoff becomes more demanding where the first-order MI approximation is also less reliable. Alternative rejection is measured at the common nominal 5% threshold. Because the methods have different null rejection rates, the curves should not be read as a comparison at matched achieved false-positive rates. They show the practical trade-off under the procedures as implemented: in these examples, the extra caution reduces detection as well as false positives.

6.3 Validity under sparse sampling

An observed table can omit many cells even when every population cell has positive probability. Figure 6.6 pushes the margins toward one dominant category while holding $n_P = n_Q = 100$. It asks whether near-zero rejection in very sparse tables means good calibration or simply a failure to calculate a test.

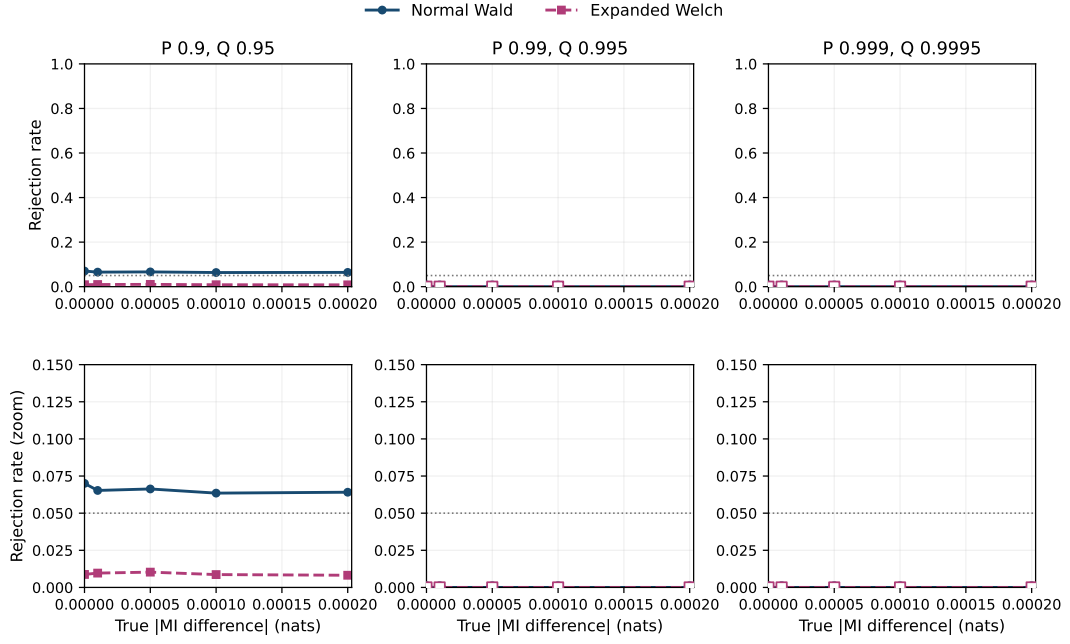


Figure 6.6: Extreme-skew 3×3 tables. Across columns the dominant row and column probabilities for (P, Q) are $(0.9, 0.95)$, $(0.99, 0.995)$, and $(0.999, 0.9995)$. Each remaining margin entry splits the leftover probability equally between the two other categories. Both use first-cell changes, $n_P = n_Q = 100$, $I(P) = 0.00001$, and $I(Q) = I(P) + |\Delta|$, for $|\Delta| = 0, 0.00001, 0.00005, 0.0001, 0.0002$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates; hollow markers flag validity below 90%.

Interpretation. For dominant margins 0.9 and 0.95, the null rates are 0.0701 for Wald and 0.0088 for Expanded Welch: the adjustment moves an excessive rate past 0.05. At the most extreme margins, 0.999 and 0.9995, both rates are zero, but only 0.0118 of Wald and 0.00005 of Expanded Welch null calculations are valid. Rejection does not meaningfully grow over the displayed alternatives there. The hollow markers show why near-zero unconditional rejection cannot be read as good calibration in sparse data.

Low validity explains much of the zero rejection in the most extreme panel. There is a second, distinct small-sample effect: Expanded Welch rejects in none of the 12 main-grid null regimes with $n = 5$, including uniform 3×3 , 5×5 and 8×8 regimes where most calculations are valid. Thus its zero rejection cannot always be explained by missing outputs. In one 8×8 different-skew alternative at $n = 5$, Wald rejects 0.30865 and Expanded Welch rejects 0.00000, while their valid-result rates are 0.83840 and 0.31415. Rejection and validity must be reported together.

Wald needs a positive usable total estimated variance. The expanded implementation additionally requires positive finite component sensitivities and degrees of freedom. Some count tables therefore produce a Wald p-value but no Expanded Welch p-value. The primary rate counts such failures as non-rejections for the procedure; they are not evidence that the population MIs are equal. The exact software rule is recorded in Appendix F.

6.4 Checks on population and sampling choices

The preceding figures isolate the central trade-off. Additional matched checks ask whether it depends on where the association sits, how the populations are constructed, the sample allocation or the table shape. Their plots and exact specifications are collected in Appendix D. Moving association between early, rare and spread cells changes the underlying table even when margins and MI are fixed (Figure D.10). An ordinal log-linear construction also changes rates under matched margins and MI (Figure D.14). Reversing which population has the larger sample or higher MI changes the alternative curves (Figures D.11 and D.12). Rectangular tables show that the conservative pattern is not exclusive to square shapes (Figure D.13). These checks limit how far any single plotted example can be generalised; none removes the need to judge null calibration and validity for the particular population pair.

The exact-independence check in Figure D.15 is kept with those supporting plots because it addresses a different mathematical boundary: the first-order MI variance becomes zero at independence. Appendix E explains why a one-table independence test has a different reference distribution.

6.5 Limits of the Student reference

Both methods use the same estimated MI difference and the same plug-in standard error. Changing the reference distribution leaves errors in either quantity untouched.

In larger different-skew null tables, the two populations can also have different remaining finite-sample MI biases even though their leading equal-sample bias corrections cancel. That can move the numerator away from zero; the present plots do not isolate how much of the excess rejection it explains.

A supplementary diagnostic makes the shared-standard-error issue concrete. For a 3×3 different-skew null with $I(P) = I(Q) = 0.02$ and $n = 100$, ordinary Wald rejects 0.0166 of independently simulated table pairs. Keeping the same estimated MI differences but dividing each by one fixed finite-sample standard deviation estimated from an independent pilot gives 0.0551. Using the fixed population first-order standard deviation instead gives 0.0976. The confirmatory Wald rate is 0.0171; its small difference from 0.0166 is Monte Carlo variation between runs.

Across the evaluated pairs, the average squared plug-in standard error is about 1.17 times the finite-sample variance of the MI difference estimated in the pilot. However, fixing the denominator changes two things at once: its size no longer varies across samples, and its relationship with the estimated MI difference is removed. The rejection-rate contrast therefore cannot be divided into separate causal shares. Appendix A.5 shows a positive first-order covariance between each table's MI estimate and its variance estimate for small positive MI on the fixed-margin path. The relationship for the two-sample statistic depends on both populations. A Student cutoff based on variance fluctuations alone does not model that joint behaviour or make the statistic exactly Student distributed.

This explains why the negative result is informative. The correction addresses uncertainty in one estimated variance, but some important finite-sample errors arise elsewhere in the shared test statistic. The fixed-denominator calculations are diagnostics, not proposed replacement tests. The supplementary results report contains their full pilot records.

6.6 Larger samples and computational cost

Figure 6.7 keeps each population pair fixed under equal MI and increases the sample size. This tests whether the poor null rates are temporary small-sample effects for the chosen populations, and whether the answer changes with their common MI.

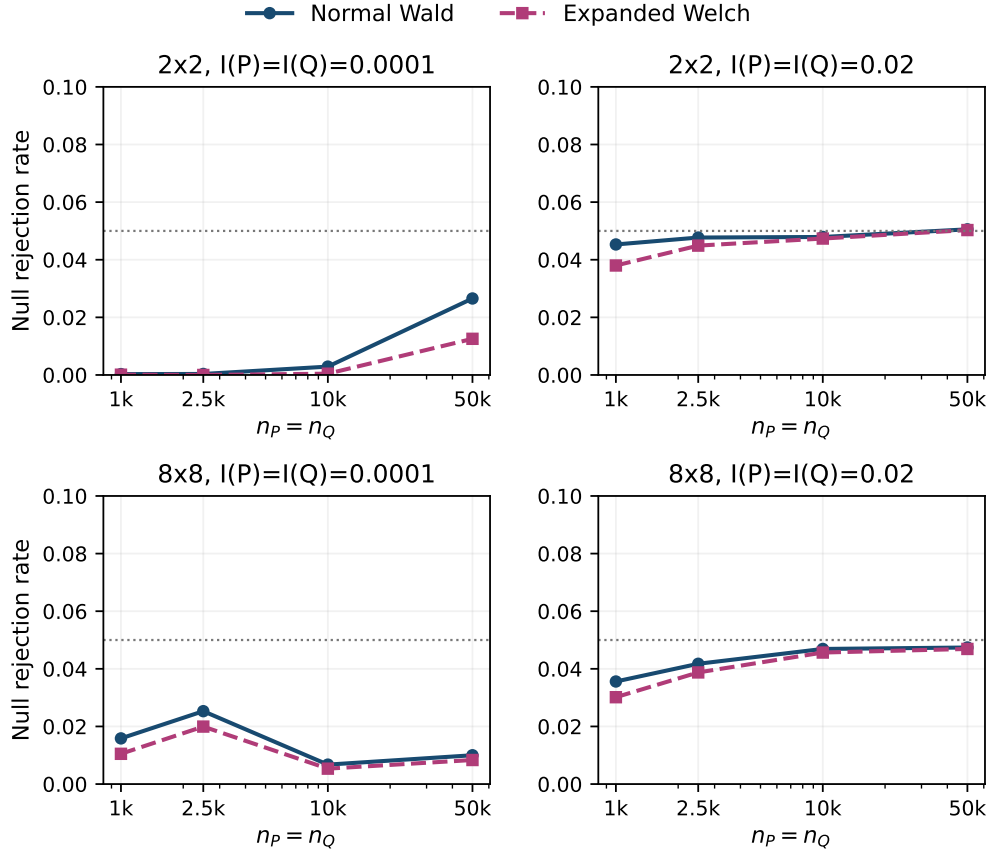


Figure 6.7: Large-sample null behaviour. Rows compare 2×2 and 8×8 tables; columns have common true MI 0.0001 and 0.02 nats. For both shapes, P 's row and column margins are $m_k(0.7)$ and Q 's are $m_k(0.8)$, with $k = 2$ or 8 . Both populations use the first-cell block. The horizontal axis is the equal sample size $n_P = n_Q = 1,000, 2,500, 10,000, 50,000$ on a labelled log scale; the vertical axis is null rejection from 0 to 0.10. Each point has 20,000 replicates.

Interpretation. The horizontal axis is sample size, and every point is a zero-difference null; there are no alternative-detection points. All displayed calculations are valid. In the displayed 2×2 and 8×8 settings, rates move toward 0.05 more clearly when common MI is 0.02 than when it is 0.0001. Across all 12 first-block shape–margin combinations at $n = 50,000$, the 0.02-MI null rates range from 0.0470 to 0.0527 for Wald and 0.0466 to 0.0526 for Expanded Welch. At MI 0.0001, the ranges remain 0.0089–0.0280 and 0.0076–0.0138. Convergence can therefore be slow near independence.

The slow approach to 0.05 at the smaller MI is consistent with the first-order approximation becoming useful more slowly near independence. These fixed populations do not establish a universal sample-size threshold for other tables.

Expanded Welch also requires more computation. A full method call has the same $O(rc)$ asymptotic complexity as Wald, but its extra sensitivity and degree-of-freedom calculations take longer in the measured implementation. Table 6.2 shows representative timings; all 60 regimes are indexed in Appendix C.

Table 6.2: Scalar-call times in microseconds for the different-skew null at $n_P = n_Q = 100$ and $I(P) = I(Q) = 0.02$. Entries give median [interquartile range] over five passes on 200 saved inputs per regime. Population construction and multinomial sampling are excluded.

Shape	Normal Wald	Expanded Welch	Expanded/Wald
2×2	85.8 [85.5, 86.3]	150.0 [149.4, 151.3]	1.75
3×3	85.1 [84.6, 85.6]	150.0 [148.9, 150.9]	1.76
3×5	85.2 [84.9, 85.5]	150.1 [149.5, 150.9]	1.76
5×5	85.3 [85.1, 85.7]	149.2 [148.5, 150.1]	1.75
8×8	88.0 [87.1, 88.9]	153.4 [152.4, 154.8]	1.74

Across the 60 timing regimes, the median call time is about 85 microseconds for Wald and 150 for Expanded Welch, a ratio of about 1.75. These timings measure calls after tables are available; they exclude population construction and sampling. The extra cost is modest beside the statistical issue. Expanded Welch is useful where Wald is demonstrably too liberal and Expanded Welch remains valid, but the figures do not support replacing Wald across all regimes. Chapter 8 states the contribution and limits of that conclusion.

Chapter 7

Implications and Limitations

Chapter 6 interprets the figures as each question is answered. This chapter draws together what those results mean for using the test and how far the conclusions can be taken.

7.1 Implications for using Expanded Welch

Normal Wald and Expanded Welch use the same estimated MI difference and standard error. Expanded Welch can only make a valid two-sided p-value larger. This is helpful when Wald rejects an equal-MI null too often; it is harmful when Wald already rejects too rarely. Its extra component validity requirements can also reduce the number of samples for which a p-value is available. Lower rejection is therefore not by itself evidence of better calibration.

The tested populations include equal-MI pairs with different margins, so the comparison goes beyond the special case $P = Q$. They show that table size, margin shape, starting MI and sampling allocation can change the trade-off. No single numerical adjustment worked well across all those settings. For applications resembling a tested regime, Normal Wald remains the simpler baseline. Expanded Welch can be a conservative sensitivity check if a separate calibration study shows that Wald is too liberal and the expanded calculation is usually valid. Choosing the method after seeing which p-value is more favourable would not be a calibrated test.

The extra computation is not the principal obstacle. In the measured implementation, an Expanded Welch call takes about 1.75 times as long as a Wald call, but both have the same $O(rc)$ operation growth for an $r \times c$ table. The statistical limitations matter more than this timing difference.

7.2 Limits of the evidence

The study evaluates fixed, deliberately constructed population pairs, not a random sample from every possible joint distribution. Its 20,000 replicates per configuration make the rate for each fixed pair precise; they do not make the selected pairs representative of every application. The main construction changes four cells. Rare-cell, spread and log-linear constructions test other patterns, but the study does not pair equal-MI populations whose association is placed in different cells. That is a specific remaining design limitation.

The methods are studied with two independent samples from fixed categorical alphabets, a two-sided 5% threshold and the stated sample ladders. The analysis does not establish performance for paired observations, continuous MI estimators, growing alphabets or a real-data workflow. Exact independence is a separate boundary because the first-order MI variance vanishes there. The dedicated check and [Appendix E](#) describe that boundary but do not turn the positive-MI test into a general independence test.

The protocol was fixed before the final simulation, following exploratory development; it was not externally preregistered. Some configurations appear in more than one figure, but each was simulated once. Repeated display is not independent confirmation, and the pointwise Monte Carlo intervals do not quantify uncertainty about untested populations.

These limits point to two distinct next steps: studying the joint finite-sample behaviour of the estimated MI difference and its standard error, and validating any alternative test under the equal-MI null when $P \neq Q$. A method for the stronger null $P = Q$ cannot be adopted for that question without checking its assumptions.

Chapter 8

Conclusion

This thesis studied how to compare the mutual information (MI) of two independently sampled categorical populations. Equal MI permits the population tables, their margins and their patterns of association to differ. The work makes two contributions to this problem. First, it derives how a change in one observation changes the plug-in estimate of MI variance, including the effects on cell probabilities and row and column totals. This gives an MI-specific Welch–Satterthwaite degrees-of-freedom calculation. Second, it evaluates that calculation against Normal Wald on fixed populations whose true MI differences are known, reporting false positives, detection, validity and runtime together.

The main finding is a qualified negative result. Expanded Welch reduces rejection relative to Wald and can move an excessive false-positive rate toward 5%. The same change makes an already-low rate lower and often reduces detection when MI differs. In sparse samples, it can also fail to return a p-value more often than Wald. The reference distribution alone cannot correct errors in the estimated MI difference or the standard error shared by both tests. This identifies both the value and the limit of a familiar variance-reference adjustment when applied to MI. Normal Wald remains the simpler general baseline for the tested task; Expanded Welch is best treated as a sensitivity check in a regime where its calibration and validity have been examined separately.

That recommendation is bounded by the experiment. Its population pairs were chosen to expose particular table sizes, margins, association patterns and sample sizes, not to represent all joint distributions. The derivation assumes independent observations, positive population cell probabilities and positive first-order MI variance. Exact independence requires a different approximation, and the study does not establish a rule for paired data, continuous MI estimators or growing alphabets.

The most useful next step is to study the estimated MI difference and its standard error together, especially near independence and in sparse tables. A likelihood-ratio or resampling method could provide a different comparison, but it would need validation for the actual null $I(P) = I(Q)$, including cases where $P \neq Q$. A real-data application and equal-MI pairs with association in different cells would then test how far the present simulation conclusions extend.

Appendix A

Additional Derivation Details

This appendix records algebra that supports Chapters 3 and 4 but is not required to follow the main argument.

A.1 Gradient of mutual information on the probability simplex

Mutual information can be written as

$$I(P) = \sum_{i,j} p_{ij} \log p_{ij} - \sum_i p_{i+} \log p_{i+} - \sum_j p_{+j} \log p_{+j}. \quad (\text{A.1})$$

Treating the cell probabilities as coordinates gives

$$\frac{\partial I(P)}{\partial p_{ij}} = (\log p_{ij} + 1) - (\log p_{i+} + 1) - (\log p_{+j} + 1) \quad (\text{A.2})$$

$$= \log \frac{p_{ij}}{p_{i+} p_{+j}} - 1 \quad (\text{A.3})$$

$$= \ell_P(i, j) - 1. \quad (\text{A.4})$$

Any valid probability perturbation $h = (h_{ij})$ satisfies $\sum_{i,j} h_{ij} = 0$. Its directional derivative is therefore

$$\left. \frac{d}{d\varepsilon} I(P + \varepsilon h) \right|_{\varepsilon=0} = \sum_{i,j} \{\ell_P(i, j) - 1\} h_{ij} \quad (\text{A.5})$$

$$= \sum_{i,j} \ell_P(i, j) h_{ij}. \quad (\text{A.6})$$

The constant in the unconstrained gradient disappears on the simplex. For the one-cell direction $h_{ij} = \mathbf{1}\{i = x, j = y\} - p_{ij}$, Equation (A.6) becomes

$$\ell_P(x, y) - \sum_{i,j} p_{ij} \ell_P(i, j) = \ell_P(x, y) - I(P), \quad (\text{A.7})$$

which is Equation (4.18).

A.2 Why the variance of pointwise MI appears

For empirical cell probabilities,

$$\hat{p}_{ij} - p_{ij} = \frac{1}{n_P} \sum_{a=1}^{n_P} \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\}. \quad (\text{A.8})$$

Substitution into the first-order Taylor expansion of MI around P gives

$$\hat{I}(P) - I(P) \approx \sum_{i,j} \ell_P(i, j) (\hat{p}_{ij} - p_{ij}) \quad (\text{A.9})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \left[\sum_{i,j} \ell_P(i, j) \{\mathbf{1}(Z_a^{(P)} = (i, j)) - p_{ij}\} \right] \quad (\text{A.10})$$

$$= \frac{1}{n_P} \sum_{a=1}^{n_P} \{\ell_P(Z_a^{(P)}) - I(P)\}. \quad (\text{A.11})$$

The observation-level summand has variance

$$\text{Var}_P\{\ell_P(X, Y) - I(P)\} = \mathbb{E}_P \left[\{\ell_P(X, Y) - I(P)\}^2 \right] \quad (\text{A.12})$$

$$= \sum_{i,j} p_{ij} \{\ell_P(i, j) - I(P)\}^2 \quad (\text{A.13})$$

$$= V(P). \quad (\text{A.14})$$

This is why the first-order variance of the plug-in MI estimator is $V(P)/n_P$, not a variance of the numerical category labels.

A.3 Mean of the variance influence function

The influence function $g_P(x, y)$ is centred at the population level. This can be shown without expanding its individual terms. The probability-weighted average of

the one-cell perturbation directions is

$$\sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\} = P - P = 0. \quad (\text{A.15})$$

The first derivative is linear in the direction of change. Therefore, the weighted average of the one-cell derivatives is the derivative in the weighted-average direction:

$$E_P\{g_P(X, Y)\} = \sum_{x,y} p_{xy} \left. \frac{d}{d\varepsilon} V\{(1 - \varepsilon)P + \varepsilon\delta_{(x,y)}\} \right|_0 \quad (\text{A.16})$$

$$= \left. \frac{d}{d\varepsilon} V\left\{P + \varepsilon \sum_{x,y} p_{xy} \{\delta_{(x,y)} - P\}\right\} \right|_0 \quad (\text{A.17})$$

$$= \left. \frac{d}{d\varepsilon} V(P) \right|_0 = 0. \quad (\text{A.18})$$

The same weighted cancellation holds after exact plug-in substitution. The implementation nevertheless subtracts the probability-weighted empirical mean inside Equation (4.38) to protect the variance calculation from floating-point roundoff; this does not change the population definition.

A.4 Direct table substitutions

For observed counts $N^P = (N_{ij}^P)$, the quantities used by expanded Welch are calculated as

$$\hat{p}_{ij} = \frac{N_{ij}^P}{n_P}, \quad \hat{p}_{i+} = \sum_j \hat{p}_{ij}, \quad \hat{p}_{+j} = \sum_i \hat{p}_{ij}, \quad (\text{A.19})$$

$$\hat{\ell}_P(i, j) = \log \frac{\hat{p}_{ij}}{\hat{p}_{i+}\hat{p}_{+j}}, \quad \hat{I}(P) = \sum_{i,j} \hat{p}_{ij} \hat{\ell}_P(i, j), \quad (\text{A.20})$$

$$\hat{V}(P) = \sum_{i,j} \hat{p}_{ij} \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2. \quad (\text{A.21})$$

All sums involving pointwise MI are restricted to cells with $\hat{p}_{ij} > 0$. Direct substitution gives

$$\hat{g}_P(i, j) = \{\hat{\ell}_P(i, j) - \hat{I}(P)\}^2 - \hat{V}(P) \quad (\text{A.22})$$

$$+ 2 \left\{ \hat{\ell}_P(i, j) - \frac{\sum_{j'} \hat{p}_{ij'} \hat{\ell}_P(i, j')}{\hat{p}_{i+}} - \frac{\sum_{i'} \hat{p}_{i'j} \hat{\ell}_P(i', j)}{\hat{p}_{+j}} + \hat{I}(P) \right\}, \quad (\text{A.23})$$

$$\hat{\tau}^2(P) = \sum_{i,j} \hat{p}_{ij} \left\{ \hat{g}_P(i, j) - \sum_{k,l} \hat{p}_{kl} \hat{g}_P(k, l) \right\}^2, \quad (\text{A.24})$$

$$\hat{\nu}_P = \frac{2n_P \hat{V}(P)^2}{\hat{\tau}^2(P)}. \quad (\text{A.25})$$

These expressions are the direct $O(rc)$ implementation of the population derivation.

A.5 Dependence between the MI and variance estimates

Both $\hat{I}(P)$ and $\hat{V}(P)$ are calculated from the same sample. Under fixed positive support, their first-order covariance is

$$\text{Cov}\{\hat{I}(P), \hat{V}(P)\} \approx \frac{1}{n_P} \mathbb{E}_P[\{\ell_P(X, Y) - I(P)\}g_P(X, Y)]. \quad (\text{A.26})$$

The corresponding expression for Q replaces P and n_P with Q and n_Q . To see its direction near independence, use the fixed-margin path $P_t = (q_{ij} + tH_{ij})$ from Section 4.5, where $q_{ij} = a_i b_j > 0$ and H has zero row and column sums. For small positive t , the expansions in that section give

$$\ell_{P_t}(i, j) - I(P_t) = \frac{tH_{ij}}{q_{ij}} + O(t^2), \quad g_{P_t}(i, j) = \frac{2tH_{ij}}{q_{ij}} + O(t^2). \quad (\text{A.27})$$

Substituting both expressions into the first-order covariance above gives

$$\text{Cov}\{\hat{I}(P_t), \hat{V}(P_t)\} \approx \frac{2t^2}{n_P} \sum_{i,j} \frac{H_{ij}^2}{q_{ij}} + O\left(\frac{t^3}{n_P}\right) \quad (\text{A.28})$$

$$= \frac{2V(P_t)}{n_P} + O\left(\frac{t^3}{n_P}\right). \quad (\text{A.29})$$

Thus the first-order large-sample approximation gives positive component covariance for small positive MI along this path. This is not a finite-sample guarantee or a result for exact independence: at $t = 0$, $V(P_0) = 0$ and the first-order MI approximation no longer describes its sampling variation. Nor does the component covariance alone determine the two-sample test's rejection rate.

A.6 Reference ordering and zero components

For positive component degrees of freedom, substituting Equation (4.39) and its Q analogue into Equation (4.40) gives the equivalent combined form

$$\hat{\nu}_{\text{expanded}} = \frac{\{\hat{V}(P)/n_P + \hat{V}(Q)/n_Q\}^2}{\hat{\tau}^2(P)/(2n_P^3) + \hat{\tau}^2(Q)/(2n_Q^3)}. \quad (\text{A.30})$$

This direct form remains defined if one table has $\hat{V} = \hat{\tau}^2 = 0$ and the other supplies a positive denominator contribution. The zero table drops out, leaving the other table's component degree of freedom. The evaluated implementation instead requires both component degrees of freedom to be positive and finite before combining them. It records such a one-zero-component case as invalid, even though the direct form has a value.

The Student p-value is no smaller than the Normal Wald p-value for the same finite statistic. To see this, write a Student variable as $Z/\sqrt{S}$, where $Z \sim N(0, 1)$, $S \sim \chi_\nu^2/\nu$, and Z and S are independent. For $c > 0$, the conditional two-sided tail $2\{1 - \Phi(c\sqrt{s})\}$ is strictly convex in s , since its second derivative is $c\phi(c\sqrt{s})(s^{-3/2} + c^2s^{-1/2})/2 > 0$. As $\mathbb{E}S = 1$, Jensen's inequality gives $\Pr(|Z/\sqrt{S}| > c) > \Pr(|Z| > c)$; at $c = 0$ both tails equal one. Moreover, the implemented Expanded-valid set is a subset of the Wald-valid set. At a common fixed threshold, every Expanded rejection is therefore also a Wald rejection, both replicate by replicate and in unconditional rates that count invalid outputs as non-rejections.

A.7 Numerical derivative checks

The implementation tests the analytic derivatives against centred finite differences on positive probability tables. For a selected cell and a small positive step η , define

$$P_{\pm\eta} = (1 \mp \eta)P \pm \eta\delta_{(x,y)}. \quad (\text{A.31})$$

For sufficiently small η , the checks compare

$$\frac{I(P_\eta) - I(P_{-\eta})}{2\eta} \quad \text{with} \quad \ell_P(x, y) - I(P), \quad (\text{A.32})$$

$$\frac{V(P_\eta) - V(P_{-\eta})}{2\eta} \quad \text{with} \quad g_P(x, y). \quad (\text{A.33})$$

The perturbations preserve positive probabilities. These checks verify the algebra independently of the Monte Carlo experiment.

Appendix B

Population Construction and Exact Design Settings

The experiment uses fixed population tables. This appendix states the construction precisely enough to reproduce any plotted point and records the finite parameter grids. Full-precision probabilities for every constructed pair are stored in `population_definitions.csv`; rounded examples in the thesis are for reading, not simulation input.

B.1 Additive construction

For prescribed positive row vector a and column vector b , form $B = ab^\top$. The principal first-cell direction H contains $\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}$ in rows and columns 1 and 2 and zero elsewhere. The rare-cell direction places the same block in the last two rows and columns. The spread direction is

$$H_{ij} = u_i v_j, \quad u_i = -1 + \frac{2(i-1)}{r-1}, \quad v_j = -1 + \frac{2(j-1)}{c-1}. \quad (\text{B.1})$$

Both score vectors sum to zero, so every direction preserves margins.

For $t \geq 0$, define $R(t) = B + tH$. Calculate the first strength at which a decreasing cell reaches zero,

$$t_{\max} = \min_{H_{ij} < 0} \frac{B_{ij}}{-H_{ij}}. \quad (\text{B.2})$$

Evaluate the path endpoint MI. A target MI must lie strictly below this endpoint, within the stated numerical tolerance. For a positive target, Brent's one-dimensional root search ([Brent, 1973](#)) solves $I\{R(t)\} = I_{\text{target}}$ on $[0, t_{\max}]$; a zero target uses

$t = 0$. The final table is accepted only if every cell is strictly positive, the sum is one, both margins agree with their prescribed vectors, and achieved MI agrees with the target. The normalisation and margin tolerances are 10^{-12} ; the MI tolerance is $10^{-12} + 10^{-8}I_{\text{target}}$. These numerical checks ensure that a horizontal-axis difference denotes constructed population truth, not an empirical estimate. They do not impose an expected-count floor on the subsequent sample.

The row and column vectors are separately constructed for each population. For shape $r \times c$, profile **uniform** uses uniform vectors of lengths r and c for both P and Q . Profile **same_skew** uses $m_r(0.8)$, $m_c(0.8)$ for both. Profile **different_skew** uses $m_r(0.7)$, $m_c(0.7)$ for P and $m_r(0.8)$, $m_c(0.8)$ for Q . The extreme regimes instead state the two dominant probabilities directly. The direction of the alternative is a separate parameter and should not be inferred from which population has the more skewed margin.

The log-linear comparison uses the scores u_i, v_j from Equation (B.1), but starts from positive cell weights

$$K_{ij}(\lambda) = \exp(\lambda u_i v_j) \quad (\text{B.3})$$

instead of adding tH to the independence table B . Iterative proportional fitting (Deming and Stephan, 1940) scales rows and columns of these weights until the prescribed margins are restored. At $\lambda = 0$, this gives the independence table; a numerical search then chooses λ to reach the target MI. The final table passes the same positivity, normalisation, margin and target-MI checks as the additive table. Unlike the first-block additive construction used in the comparison, it changes cells throughout the table. Matching margins and MI does not imply matching all cell probabilities or expected counts.

B.2 Frozen statistical grids

Unless a row of Table B.1 says otherwise, both populations use the first-cell additive direction, $I(P) = I_0$, $I(Q) = I_0 + |\Delta|$, and the three standard margin profiles. The symbol n means $n_P = n_Q = n$. A requested point was simulated only after the population constructor verified a strictly positive feasible table. The final display contains no infeasible slots, but that is a result of choosing feasible finite grids. Curves should not be extrapolated beyond their last constructed target.

Table B.1: Exact statistical experiment grids; MI values are in nats.

Family	Settings
Main	Shapes $2 \times 2, 3 \times 3, 5 \times 5, 8 \times 8$; three profiles; $I_0 = 0.02$; $ \Delta \in \{0, 0.001, 0.002, 0.005, 0.01, 0.02\}$; $n \in \{2, 5, 10, 20, 50, 100, 250, 500, 1000\}$.
Baseline	Main shapes and profiles; $I_0 \in \{0.0001, 0.001, 0.02\}$; main difference grid; $n \in \{20, 100, 1000\}$.
Changed cells	Shapes $3 \times 3, 5 \times 5, 8 \times 8$; different-skew margins; $I_0 = 0.0001$; $ \Delta \in \{0, 0.0001, 0.00025, 0.0005, 0.001, 0.002\}$; $n \in \{20, 100, 1000\}$. First and rare blocks for all shapes; spread pattern for 5×5 and 8×8 .
Imbalance	Main shapes, profiles and difference grid; $I_0 = 0.02$. Sample pairs are $(50, 50), (50, 100), (100, 50), (50, 250), (250, 50), (50, 500), (500, 50), (500, 500)$. Both Q -higher and P -higher MI directions.
Larger effects	Shapes $2 \times 2, 3 \times 3$; three profiles; $I_0 = 0.02$; main difference grid plus $0.05, 0.1, 0.2$; $n \in \{20, 100, 1000\}$.
Rectangular	Shapes $2 \times 3, 3 \times 5$; three profiles; $I_0 = 0.02$; main difference grid; $n \in \{20, 100, 1000\}$.
Construction	Shapes $3 \times 3, 5 \times 5, 8 \times 8$; uniform and different-skew profiles; $I_0 \in \{0.0001, 0.02\}$; main difference grid; $n \in \{20, 100, 1000\}$; additive first block and ordinal log-linear constructors.
Extreme skew	Shapes $2 \times 2, 3 \times 3, 8 \times 8$; dominant margin probability pairs $(\rho_P, \rho_Q) \in \{(0.9, 0.95), (0.99, 0.995), (0.999, 0.9995)\}$; $I_0 = 0.00001$; $ \Delta \in \{0, 0.00001, 0.00005, 0.0001, 0.0002\}$; $n \in \{1, 2, 5, 20, 100, 1000, 10000\}$.
Rare-cell stress	Shapes $3 \times 3, 8 \times 8$; dominant pair $(0.9, 0.95)$; first and rare blocks; $I_0 = 0.00001$; $ \Delta \in \{0, 0.00001, 0.000025, 0.00005, 0.0001\}$; same n ladder as extreme skew.
Independence	Shapes $2 \times 2, 8 \times 8$; three profiles; $I_0 = 0$; main difference grid; $n \in \{2, 20, 100, 1000\}$.

Family	Settings
Convergence	Main square shapes and profiles; equal MI $I(P) = I(Q) \in \{0.0001, 0.02\}$; $n \in \{1000, 2500, 10000, 50000\}$. Additional rare-block controls use $3 \times 3, 5 \times 5, 8 \times 8$, different-skew margins and common MI 0.0001.

At $n = 1$, neither statistic is evaluable, so the result is recorded as invalid. Every unique statistical configuration has 20,000 sampled table pairs. When the same population pair and sample sizes appear in another section, the saved configuration is reused for presentation and not simulated again.

Appendix C

Evidence Index and Additional Diagnostics

Chapter 6 develops the main narrative around selected figures; Appendix D collects the other comparisons. This appendix identifies the supplementary results report and data files for every configuration. Paths are relative to the workspace root `Simulations/`. The final results directory used below is `WelchSatterthwaiteMI/results/thesis_redesign/`. These are workspace paths; Appendix F gives commands for the extracted archive, where `main.tex` and `WelchSatterthwaiteMI/` share one root.

The supplementary results report is at `WelchSatterthwaiteMI/docs/experiments/THESIS_EXPERIMENTS.md`. It contains 99 primary figures, seven calibration zooms and tables of plotted rates and validity. The thesis-specific PDFs in `Thesis Writeup/Welch MI/ figures_rewrite/` are generated from the two final CSV files named in the source script. The adjacent `figure_manifest.json` lists, for each thesis PDF, its original section, source files, configuration identifiers and 20,000-replicate denominator. The file `evidence_values.tex` is generated from exact configuration IDs for the numerical examples in Chapter 6.

For each configuration, 20,000 table pairs are sampled. The plotted rate is the number of valid rejections divided by 20,000. The method-valid conditional rate divides valid rejections by that method's valid outputs; it is undefined if there are none. The common-valid rate restricts to outputs where both methods are valid. A paired difference is based on counts of the four joint rejection outcomes for the same table pairs. These distinctions allow a reader to check whether a curve difference is primarily a changed reference or lost validity.

For the paired comparison, let D_s equal the Wald valid-rejection indicator minus the Expanded Welch valid-rejection indicator for sampled table pair s , matching the

Table C.1: Saved files and the question each answers.

File in the final results directory	Use
<code>display_manifest.csv</code>	Figure section, row, column, horizontal value and exact configuration identifier; a configuration may recur.
<code>configuration_manifest.csv</code>	Unique statistical configurations, sample sizes, target populations and simulation identifiers.
<code>population_definitions.csv</code>	Full-precision P and Q probability tables, margins, achieved MI, constructor and feasibility diagnostics for every pair.
<code>cell_results.csv</code>	One row per configuration and method: unconditional rejection, method validity, conditional and common-valid rates, Wilson intervals, expected-count and support-loss diagnostics.
<code>paired_method_results.csv</code>	Paired rejection counts, rate differences and approximate paired intervals from the same replicates.
<code>runtime_summary.csv</code>	Per-regime median and interquartile call times in microseconds, validity and Expanded-to-Wald time ratio.
<code>verification.json</code> and <code>report_verification.json</code>	Run-level population, denominator, pairing, method and report checks.

sign used in Table 6.1. Its average is $\bar{D} = (1/20,000) \sum_{s=1}^{20,000} D_s$, with approximate 95% interval

$$\bar{D} \pm 1.96 \left\{ \frac{1}{20,000(19,999)} \sum_{s=1}^{20,000} (D_s - \bar{D})^2 \right\}^{1/2}. \quad (\text{C.1})$$

This calculation preserves the within-replicate pairing. In the saved CSV, the generic field is stored as method A minus method B, with Expanded Welch listed as method A; Table 6.1 therefore reports its negative. The individual method intervals use Wilson’s binomial score construction (Wilson, 1927).

For any P population with sample size n_P , its minimum expected cell count is $\min_{ij} n_P p_{ij}$; define Q ’s analogously. No “expected minimum” of random observed counts is calculated. The saved support diagnostics include empty observed rows and columns and the fraction of zero-count cells. Such values describe why a statistic might fail, but do not constitute a universal validity threshold.

The thesis does not treat a descriptive count of regimes closer to 0.05 as a formal paired test over regimes. Wilson intervals quantify binomial Monte Carlo variation at each fixed point, and paired normal intervals quantify a specified within-configuration method difference. Neither interval family is a simultaneous assertion across all configurations. Tables sharing a population but changing n or $|\Delta|$ answer different fixed-design questions and should not be collapsed into one mean rate.

Appendix D

Additional Experimental Figures

Chapter 6 uses seven figures to develop the main argument. The figures below retain the other matched comparisons from the same simulation run. Each caption gives its population and sample settings, and each interpretation concerns that figure alone. The supplementary results report identified in Appendix C contains the remaining sample sizes and numerical results for every plotted point.

D.1 Remaining square-table comparisons

The principal design crosses four square table sizes with three margin profiles. Chapter 6 discusses the uniform 2×2 and 8×8 comparisons and the different-skew 8×8 comparison. The nine other shape–margin combinations follow in the same order as the design grid.

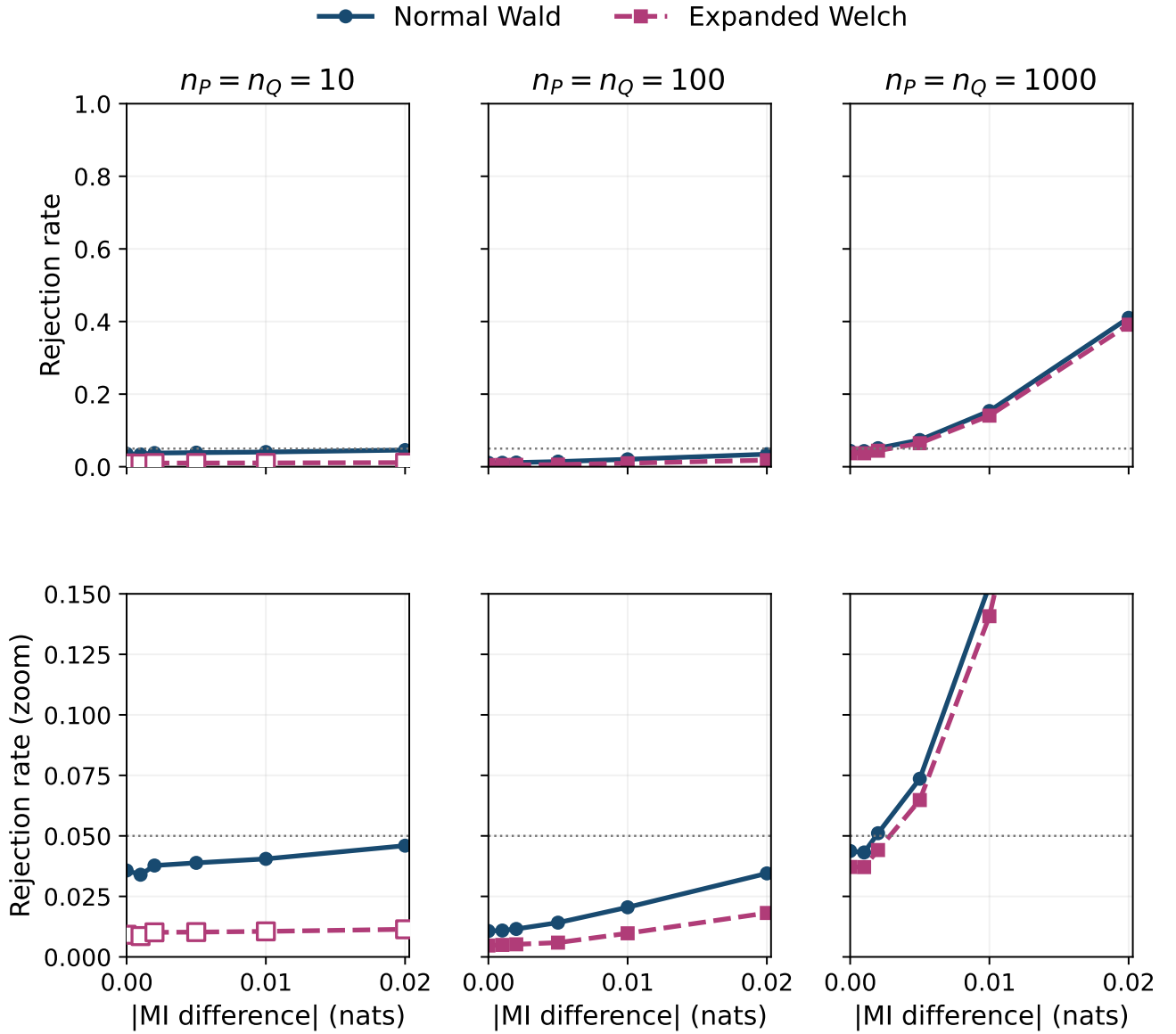


Figure D.1: 2×2 tables, both P and Q have row and column margins $m_2(0.8) = (0.8, 0.2)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. The null points are below 0.05, especially for Expanded Welch. At $n = 10$, many Expanded Welch calculations are invalid (hollow markers), so its low unconditional rate has more than one cause. Detection rises with the constructed difference mainly at $n = 1,000$; there Wald remains slightly higher. This skewed pair gives no general calibration advantage to the more conservative reference.

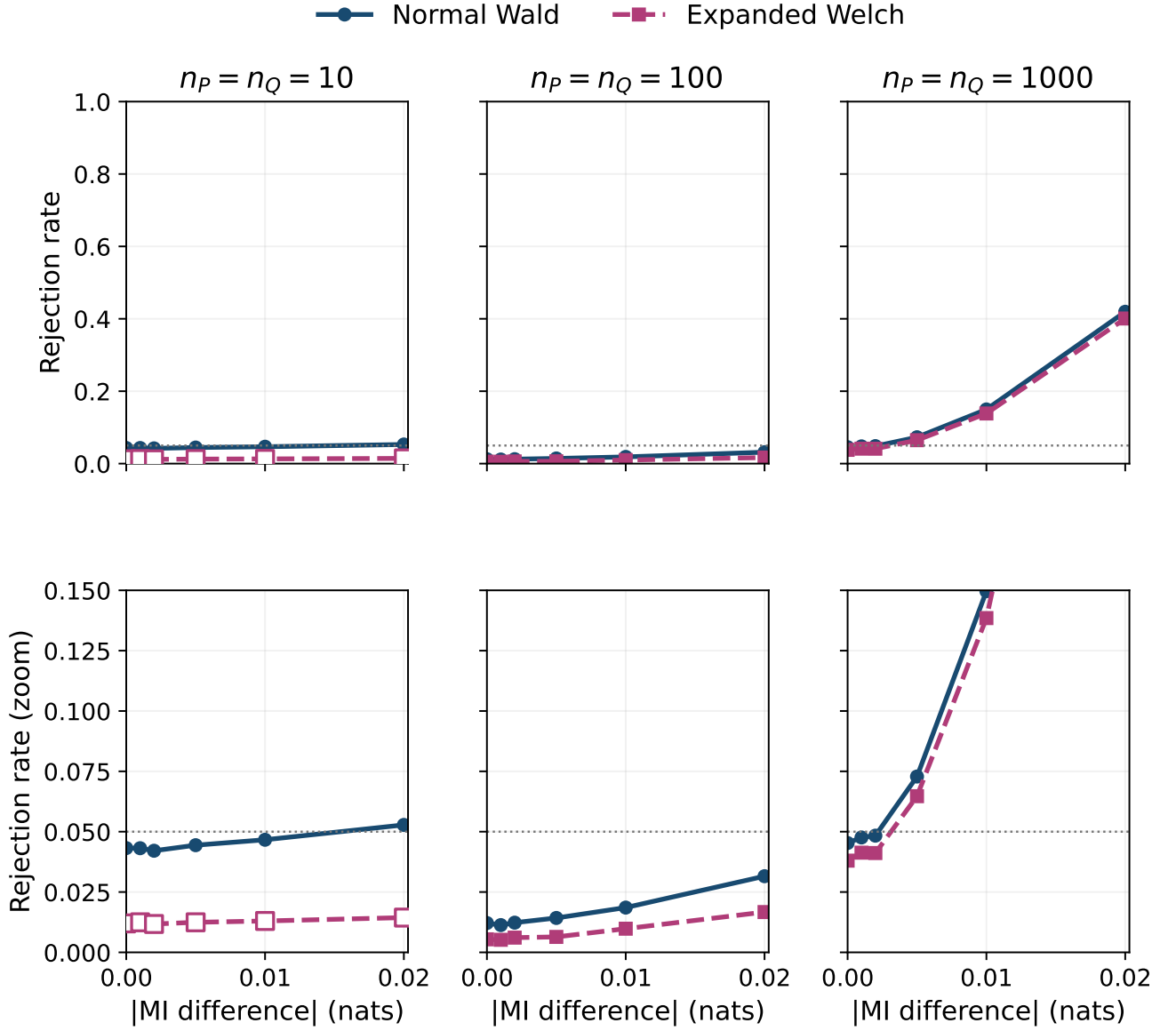


Figure D.2: 2×2 tables, P has row and column margins $m_2(0.7) = (0.7, 0.3)$, while Q has $m_2(0.8) = (0.8, 0.2)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. At zero difference, the two tables have equal MI but different margins. Both methods are mostly below the 0.05 target, and Expanded Welch is lower. At $n = 10$, its hollow markers also indicate substantial invalidity. The alternative curves rise noticeably at $n = 1,000$, with Wald just above Expanded Welch. The same conservative pattern therefore persists when the equal-MI null does not require identical populations.

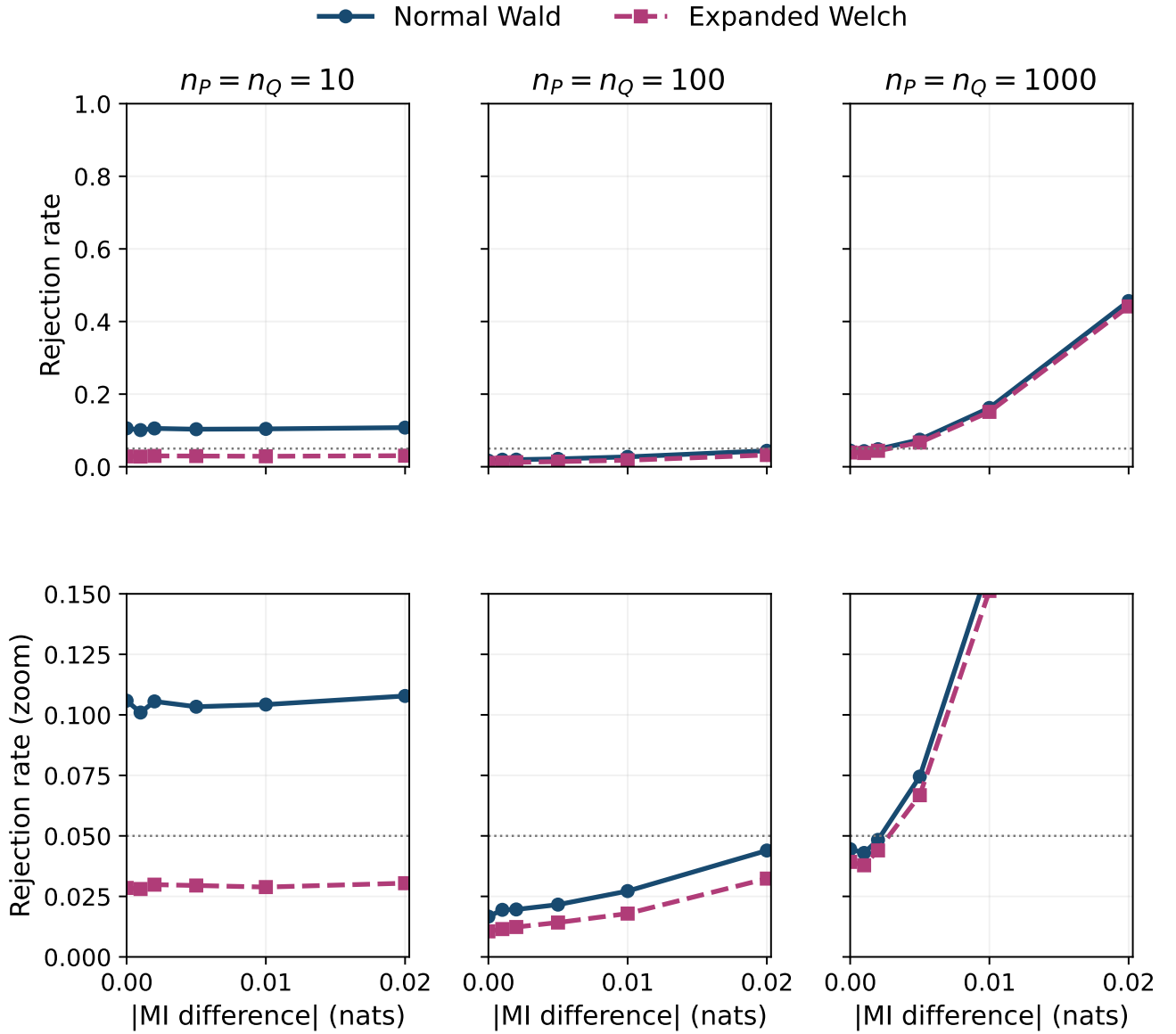


Figure D.3: 3×3 tables, both P and Q have uniform row and column margins $(1/3, 1/3, 1/3)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. At $n = 10$, Wald’s zero-difference rate is above 0.05 and Expanded Welch brings it below the line; both methods are almost always valid. At $n = 100$, however, both start well below 0.05 and the correction lowers rejection further. Detection grows strongly at $n = 1,000$, where Wald remains slightly higher. Whether the adjustment helps calibration changes with sample size, not just table shape.

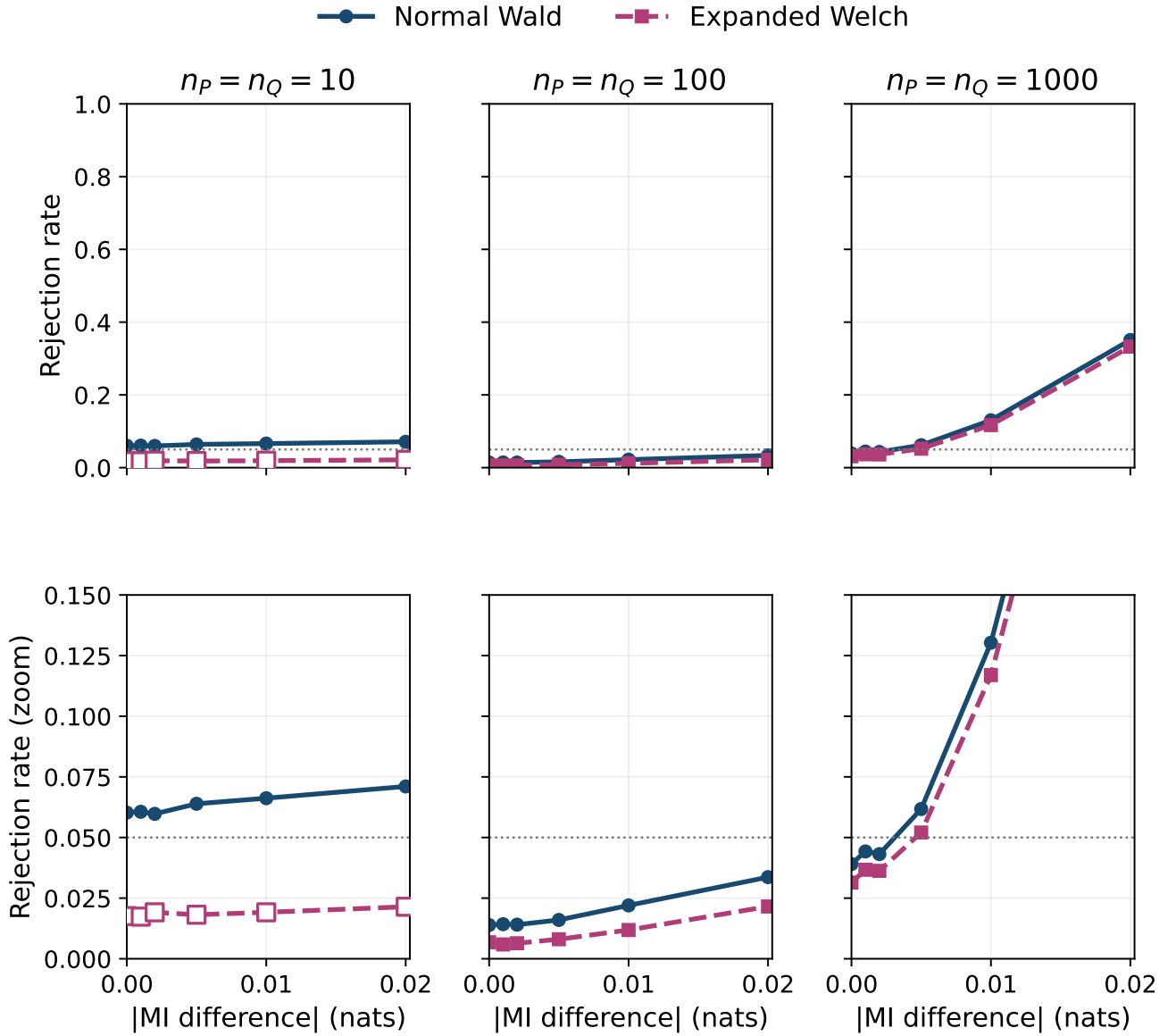


Figure D.4: 3×3 tables, both P and Q have row and column margins $m_3(0.8) = (0.8, 0.1, 0.1)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. The $n = 10$ null rate falls from slightly above 0.05 for Wald to below it for Expanded Welch, but Expanded Welch is valid in only about two thirds of those samples. At $n = 100$ and 1,000, both null rates are below 0.05. Rejection rises with the MI difference most visibly at $n = 1,000$; Expanded Welch remains lower. The small-sample null reduction is not solely a reference-distribution effect.

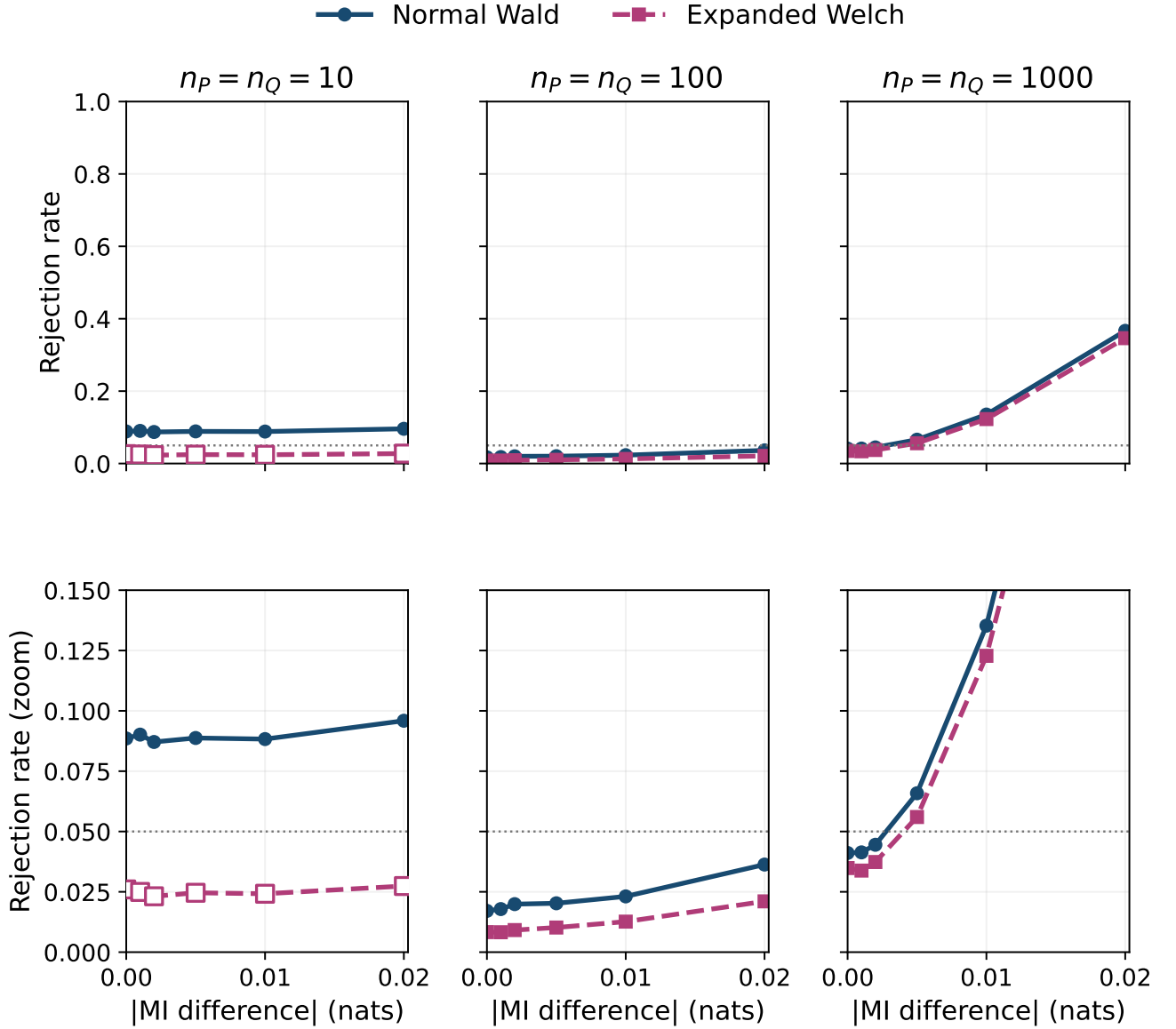


Figure D.5: 3×3 tables, P has row and column margins $m_3(0.7) = (0.7, 0.15, 0.15)$, while Q has $m_3(0.8) = (0.8, 0.1, 0.1)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. With different margins, Wald rejects too often at the $n = 10$ null; Expanded Welch rejects less, although its hollow markers show reduced validity. At $n = 100$, both are conservative. At $n = 1,000$, detection increases with the true difference and the methods draw closer. This comparison shows a small-sample benefit against liberal rejection but not a consistent improvement across sample sizes.

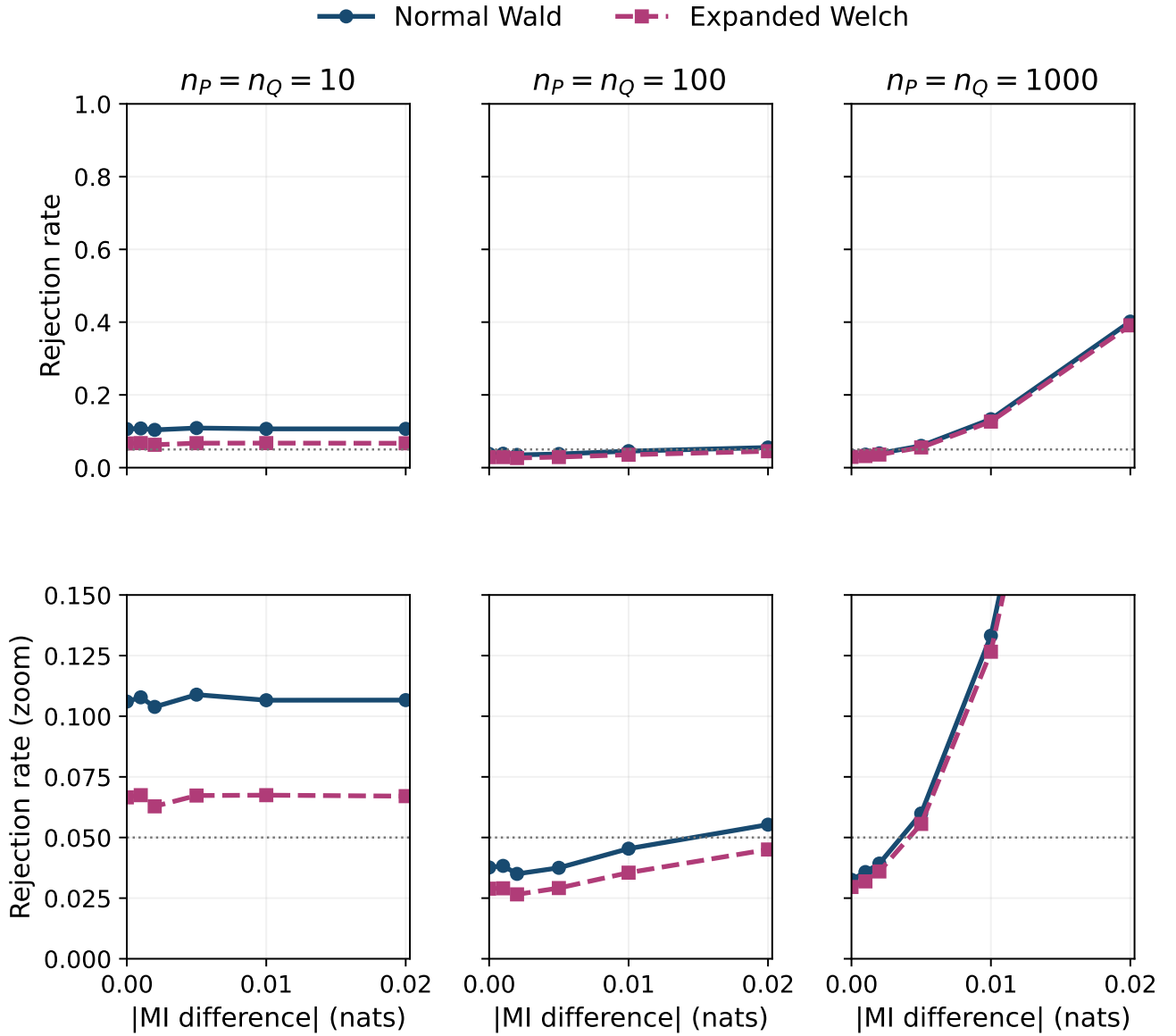


Figure D.6: 5×5 tables, both P and Q have uniform row and column margins $(1/5, \dots, 1/5)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. Both methods are valid in the displayed panels. At zero difference, Wald is liberal at $n = 10$, while Expanded Welch reduces but does not eliminate the excess. At $n = 100$ and $1,000$, both null rates are below 0.05. Little detection is gained over the displayed difference range at $n = 100$, whereas rejection grows clearly at $n = 1,000$. The correction lowers both false positives and detection.

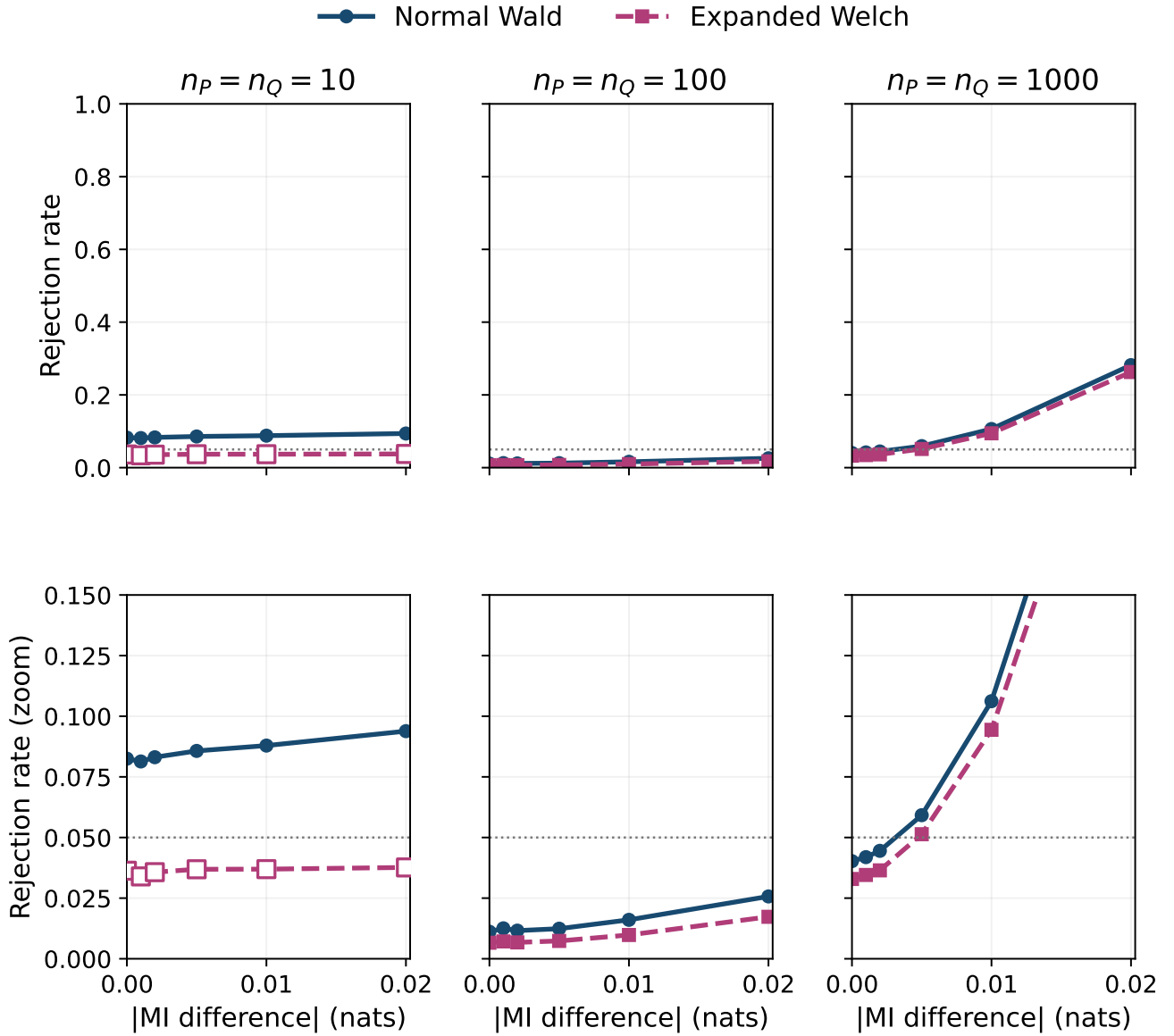


Figure D.7: 5×5 tables, both P and Q have row and column margins $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. At $n = 10$, Wald starts above the 0.05 line and Expanded Welch below it, but the latter is invalid for roughly one third of samples. At $n = 100$ and 1,000, both begin below the line. Detection increases substantially only in the largest-sample panel, with Wald higher. The small-sample appearance of better calibration must be read together with validity.

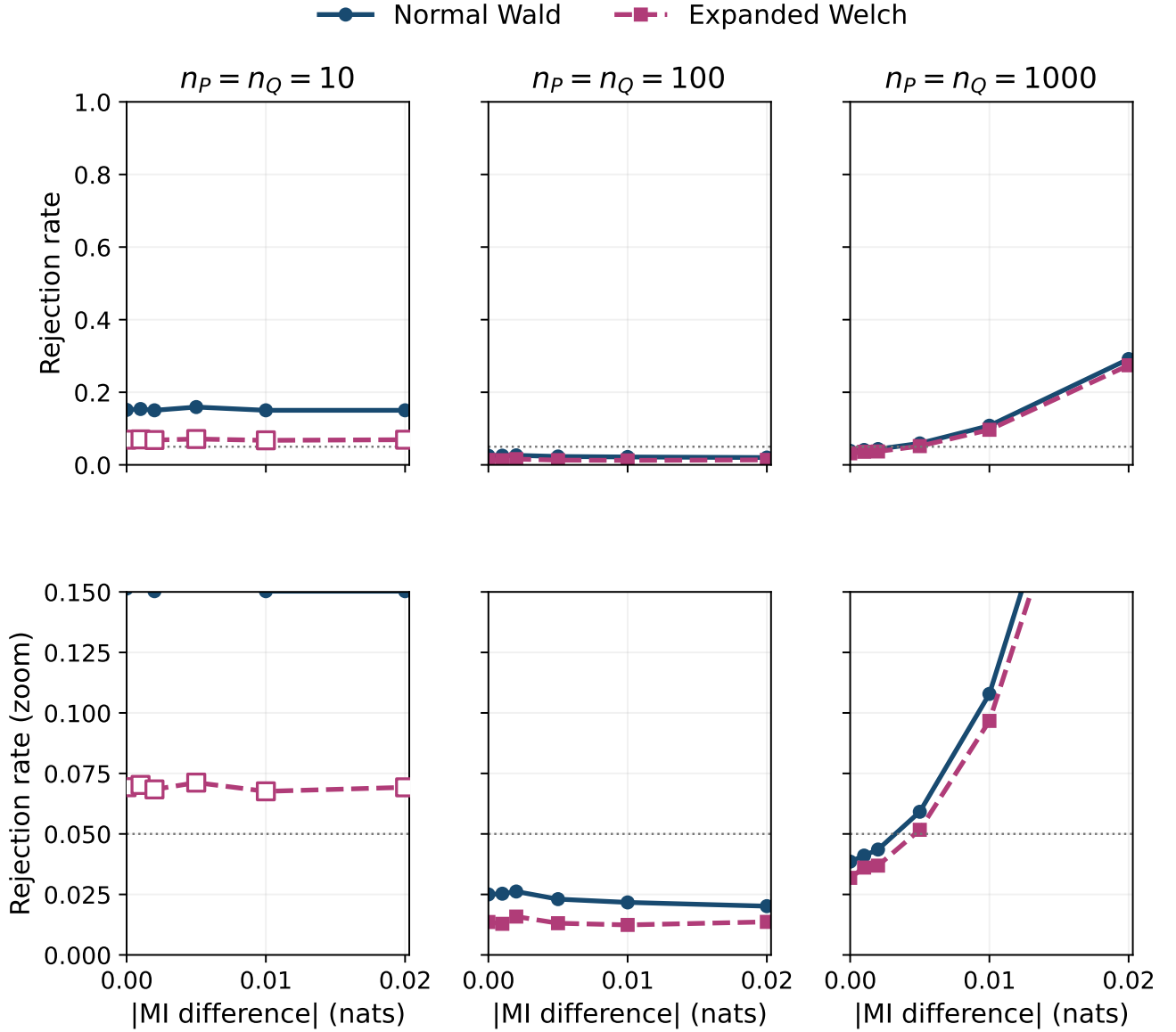


Figure D.8: 5×5 tables, P has row and column margins $m_5(0.7) = (0.7, 0.075, 0.075, 0.075, 0.075)$, while Q has $m_5(0.8) = (0.8, 0.05, 0.05, 0.05, 0.05)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. The $n = 10$ null is liberal for both methods, though less so for Expanded Welch; its hollow markers also show reduced validity. At $n = 100$, rejection stays low and even falls across this short alternative range, so these curves should not be described as increasing power. At $n = 1,000$, detection rises and Wald remains slightly higher. The benefit of reducing excess null rejection is regime-specific.

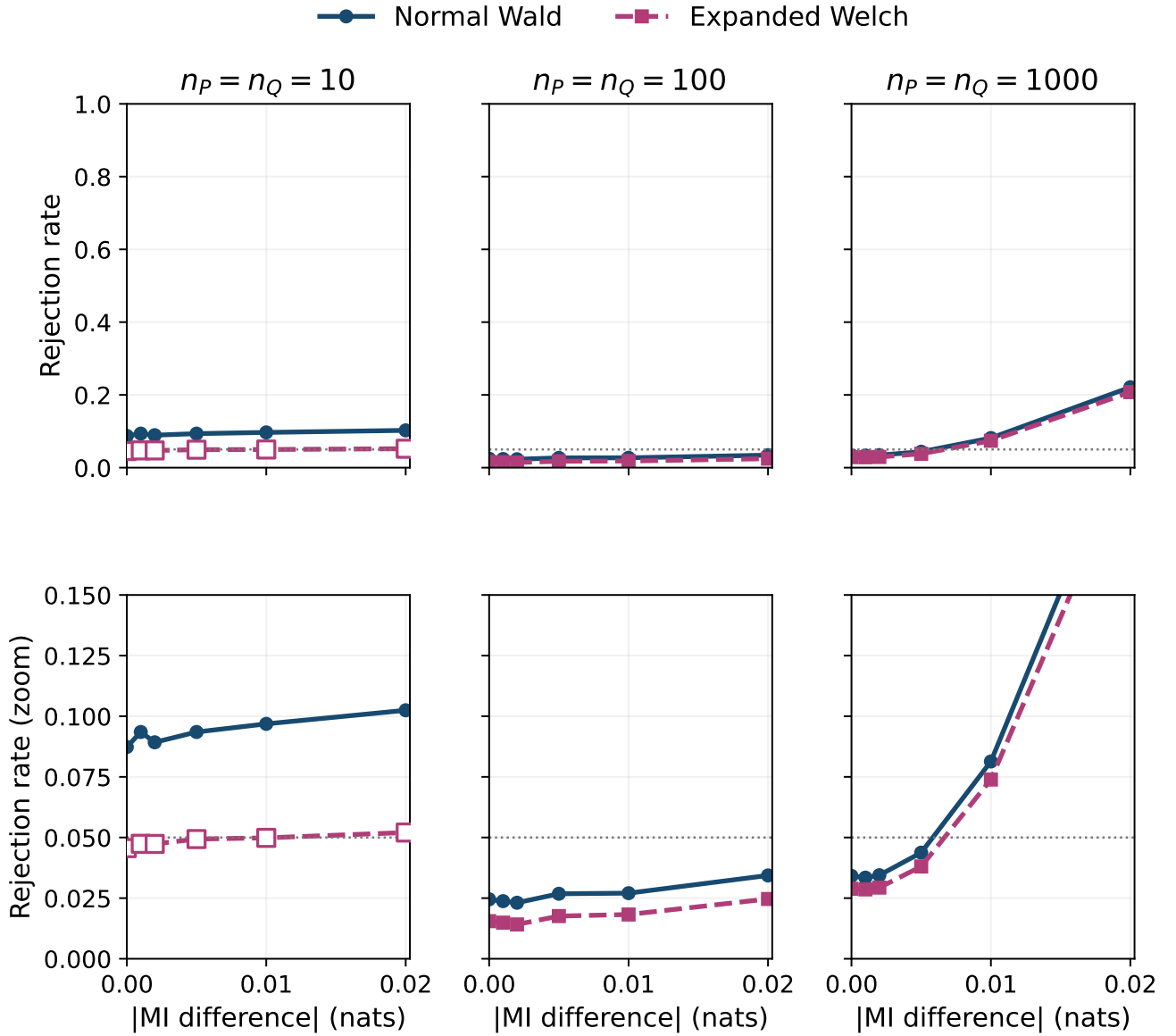


Figure D.9: 8×8 tables, both P and Q have row and column margins $m_g(0.8) = (0.8, 0.2/7, \dots, 0.2/7)$. Columns: $n_P = n_Q = 10, 100, 1000$, respectively. For each population the first two rows and columns receive the signed changes in Equation (5.3); $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$. The horizontal grid is $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. The upper panels have a shared 0–1 rejection axis and the lower panels show the same data at 0–0.15. At zero the rate is false positives; elsewhere it is detection. Each point uses 20,000 paired samples. Hollow markers denote validity below 90%.

Interpretation. At $n = 10$, Wald is above 0.05 at the null and Expanded Welch below it, but many Expanded Welch outputs are invalid. Both null rates are below 0.05 at $n = 100$ and 1,000. Across the alternative points, appreciable growth appears chiefly at $n = 1,000$, with Wald higher. The adjustment does not turn the smaller-sample curves into strong detection.

D.2 Other controlled comparisons

The following figures hold the stated margins, MI targets and sample sizes fixed while changing association location, sample allocation, table shape or population construction. The final figure starts at exact independence, where the first-order MI variance is zero.

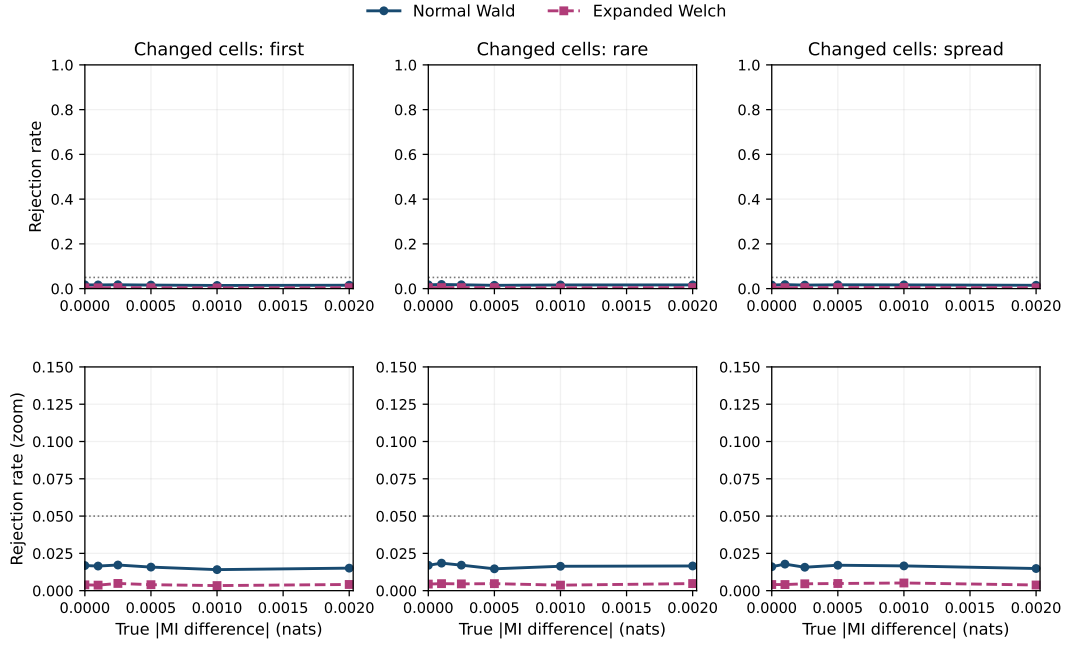


Figure D.10: Changed-cell comparison for 5×5 tables. Columns place association in the first 2×2 block, last (rare) 2×2 block, or a centred score outer product spread across all cells. For every column, P 's row and column margins are $(0.7, 0.075, 0.075, 0.075, 0.075)$, and Q 's are $(0.8, 0.05, 0.05, 0.05, 0.05)$. Here $I(P) = 0.0001$, $I(Q) = I(P) + |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.0001, 0.00025, 0.0005, 0.001, 0.002$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Interpretation. For first, rare and spread patterns, respectively, the zero-difference Wald rates are 0.0169, 0.0170 and 0.0160; Expanded Welch gives 0.0039, 0.0044 and 0.0041. All displayed calculations are valid. The alternative curves remain nearly flat over this short MI range, so neither method detects these differences well. Changing the location of association does not remove the conservative gap here, although this one plot does not make all patterns equivalent.

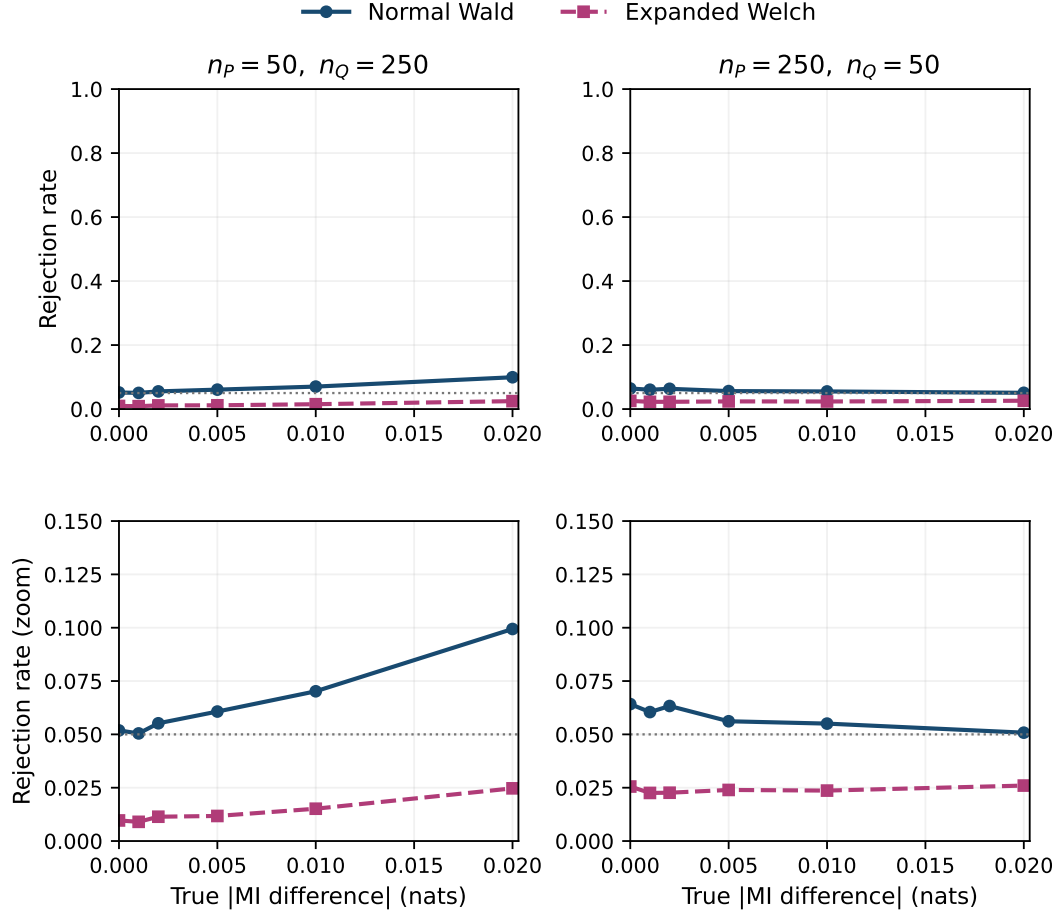


Figure D.11: Unequal samples, Q higher. Columns have $(n_P, n_Q) = (50, 250)$ and $(250, 50)$. For both columns, P has 3×3 row and column margins $(0.7, 0.15, 0.15)$ and Q has $(0.8, 0.1, 0.1)$; the first-cell change block is used. $I(P) = 0.02$ and $I(Q) = 0.02 + |\Delta|$, with $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Interpretation. At zero difference, Wald is near or above 0.05 while Expanded Welch is below it in both allocations. When Q has more observations $(n_P, n_Q) = (50, 250)$, rejection rises with $I(Q) - I(P)$; at 0.02 nats it reaches 0.0994 for Wald and 0.0247 for Expanded Welch. With P larger, the curves rise much less. These points are valid, but the alternative gap is not a size-adjusted power comparison because their matched null rates differ.

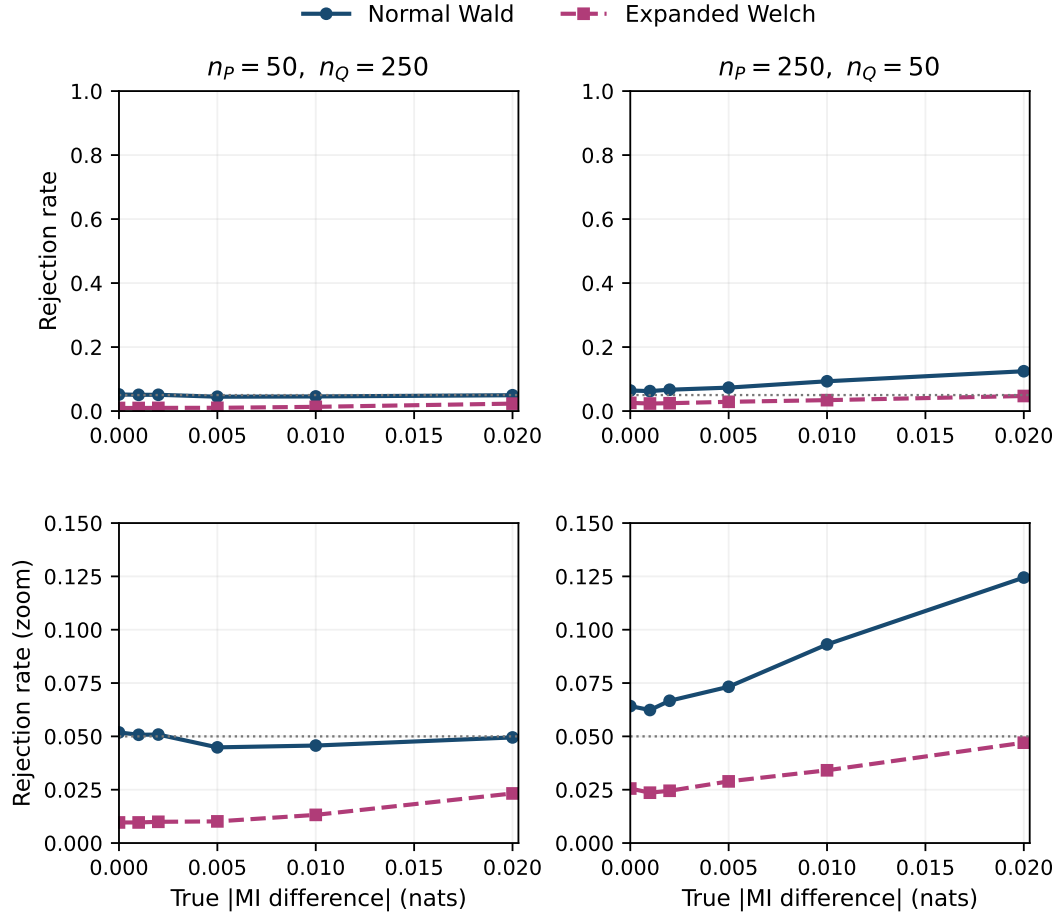


Figure D.12: Unequal samples, P higher. Columns have $(n_P, n_Q) = (50, 250)$ and $(250, 50)$. P 's 3×3 row and column margins are $(0.7, 0.15, 0.15)$; Q 's are $(0.8, 0.1, 0.1)$. Both use the first-cell change block. $I(Q) = 0.02$ and $I(P) = 0.02 + |\Delta|$, with $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Interpretation. The zero-difference rates are the same as in Figure D.11, because each allocation uses the same pair of populations in both figures at that point; P and Q still have different margins. Reversing the MI direction changes what happens next: when P has the larger sample, Wald rises well above its null rate, whereas it stays nearly flat when Q has the larger sample. At $(n_P, n_Q) = (50, 250)$ and 0.02 nats, Wald rejects 0.0495 and Expanded Welch 0.0232; both calculations are valid. Direction and allocation therefore matter even at the same absolute MI difference.

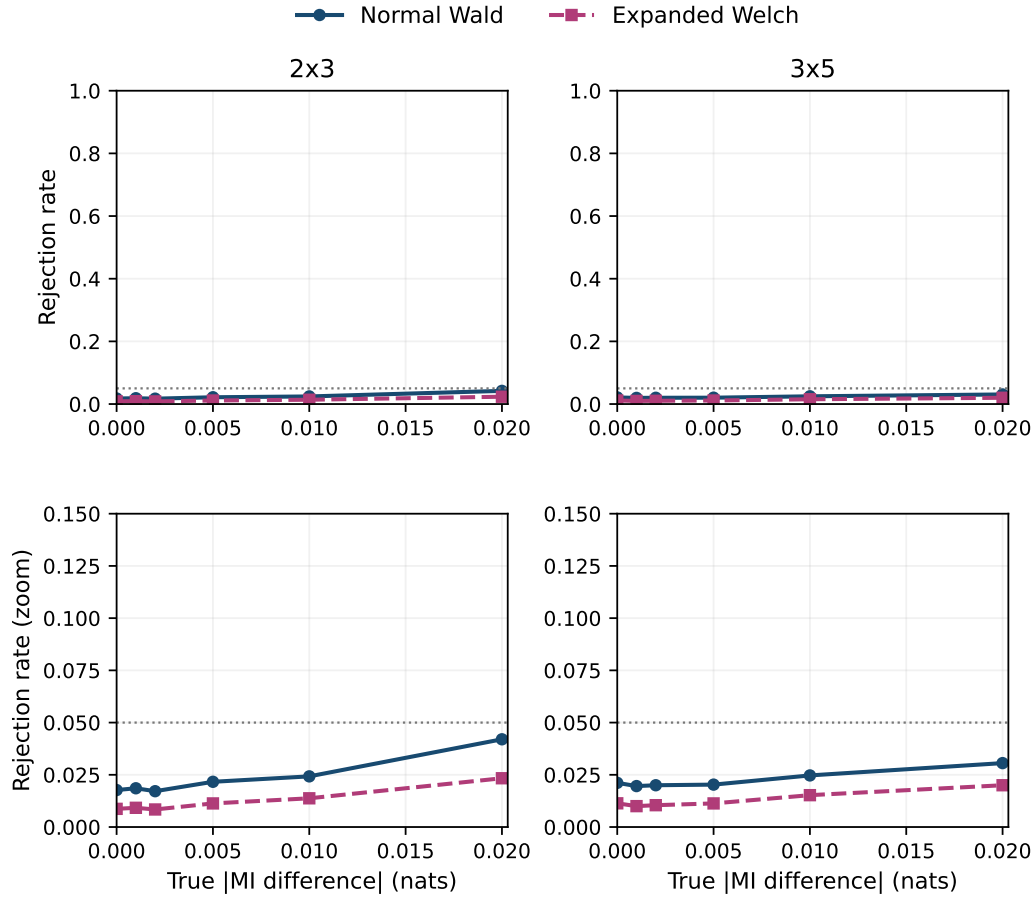


Figure D.13: Rectangular tables. Columns are 2×3 and 3×5 . In each dimension k , P 's corresponding margin is $m_k(0.7)$ and Q 's is $m_k(0.8)$, for both rows and columns. Thus the first column uses P rows $(0.7, 0.3)$, columns $(0.7, 0.15, 0.15)$, and Q rows $(0.8, 0.2)$, columns $(0.8, 0.1, 0.1)$; the second applies the same formula to 3 and 5 categories. Both use first-cell changes, $n_P = n_Q = 100$, $I(P) = 0.02$, and $I(Q) = 0.02 + |\Delta|$ for $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Interpretation. Both rectangular panels start below 0.05, with Expanded Welch lower. Rejection grows only modestly by 0.02 nats and remains below 0.05 for both methods throughout the displayed range; validity is high. Thus these shapes show weak detection over the chosen differences, not an improvement in false-positive calibration from the lower Expanded Welch curve. The conservative pattern is not confined to square tables.

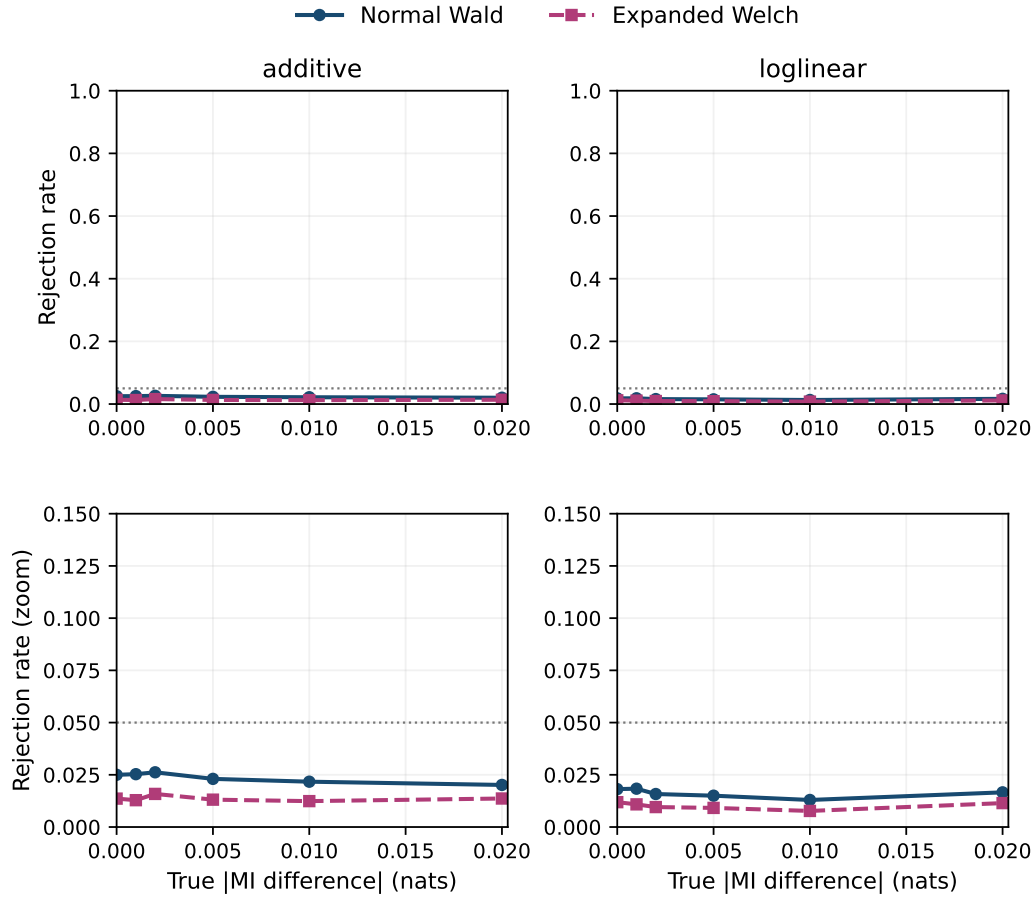


Figure D.14: Additive versus ordinal log-linear construction for 5×5 tables. The left column uses the first-cell additive block; the right fits an ordinal interaction while preserving the margins. In both, P 's row and column margins are $(0.7, 0.075, 0.075, 0.075, 0.075)$ and Q 's are $(0.8, 0.05, 0.05, 0.05, 0.05)$. Both have $I(P) = 0.02$, $I(Q) = 0.02 + |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Interpretation. At zero difference, the additive and log-linear Wald rates are 0.0250 and 0.0181; Expanded Welch gives 0.0136 and 0.0119. All four are valid and below 0.05. Rejection stays low or falls slightly as the displayed difference grows. Matching margins and MI therefore does not make finite-sample behaviour identical across constructions, but neither construction removes the conservative gap between the methods.

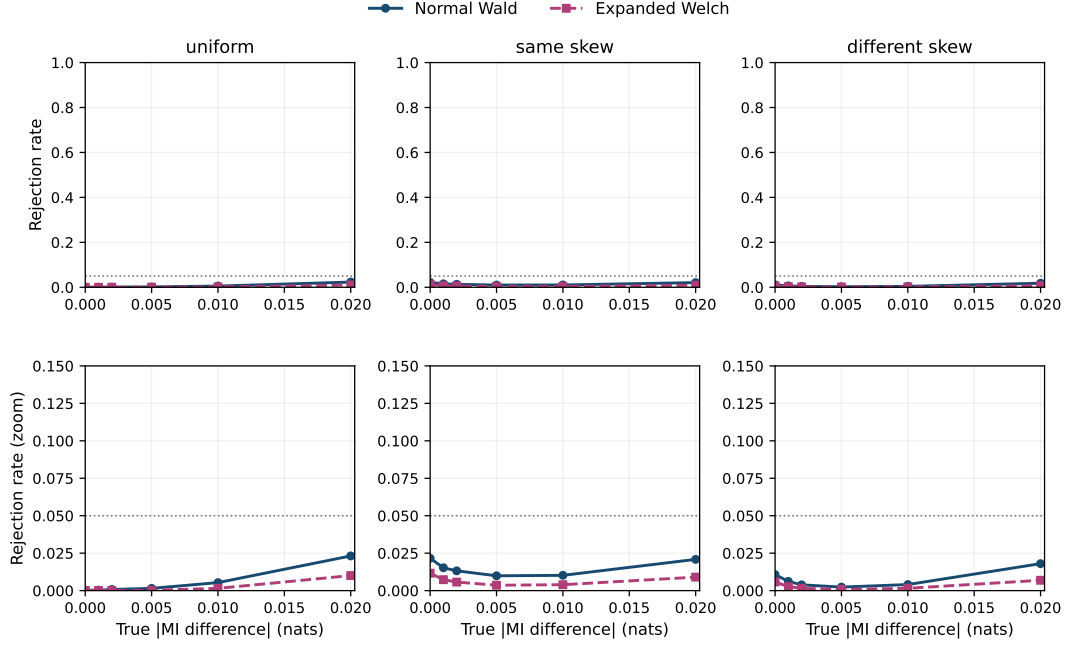


Figure D.15: Exact-independence boundary for 2×2 tables. Columns are uniform margins $(0.5, 0.5)$ for both populations, same skew $(0.8, 0.2)$ for both, and different skew with $P = (0.7, 0.3)$ and $Q = (0.8, 0.2)$; each vector is used for both rows and columns. The first-cell block is used, $I(P) = 0$, $I(Q) = |\Delta|$, $n_P = n_Q = 100$, and $|\Delta| = 0, 0.001, 0.002, 0.005, 0.01, 0.02$ nats. Top: 0–1; bottom: 0–0.15. Each point has 20,000 replicates.

Interpretation. At the uniform zero-difference point, Wald rejects 0.00025 and Expanded Welch 0.00010 of samples; validity is high. Rejection grows little over most of the displayed alternative range, and Expanded Welch remains lower. This is not a routine positive-MI validation: at exact independence the population first-order MI variance is zero, so the approximation underlying both curves changes. Appendix E explains the different reference used by a one-table independence test.

D.3 Main-grid null-rate overview

Table D.1 describes where the 108 main-grid null rates fall relative to 0.025–0.075. This interval is only a descriptive aid, not a formal calibration test. The rates and validity fractions for each configuration are available separately in the supplementary results report and saved data.

Table D.1: Location, separately for each method, of the 108 main-grid null configurations relative to the descriptive rejection-rate interval 0.025–0.075. Each profile contains 36 configurations. The final column separately counts configurations with validity below 90%; it overlaps the three rate-location columns. In particular, all four $n = 2$ configurations per profile have zero validity for both methods. Uniform and same-skew have $P = Q$; different-skew has $P \neq Q$ but $I(P) = I(Q)$. Exact rates and validity are available in the saved results described in Appendix C.

Margins	Method	Below	Inside	Above	Validity below 90%
Uniform	Normal Wald	10	18	8	4
Uniform	Expanded Welch	19	16	1	5
Same skew	Normal Wald	14	15	7	8
Same skew	Expanded Welch	25	11	0	12
Different skew	Normal Wald	9	16	11	8
Different skew	Expanded Welch	19	13	4	12

Appendix E

The Independence Boundary

The main thesis compares populations with $I(P) = I(Q) > 0$, allowing $P \neq Q$. This appendix explains why constructing an independent reference distribution does not turn the Expanded Welch method into a new first-order test of independence.

E.1 Constructing the independent reference

For a joint distribution $P = (p_{ij})$, define

$$q_{ij} = p_{i+}p_{+j}. \quad (\text{E.1})$$

Then $Q = (q_{ij})$ is a probability distribution because

$$\sum_{i,j} q_{ij} = \left(\sum_i p_{i+} \right) \left(\sum_j p_{+j} \right) = 1. \quad (\text{E.2})$$

It has the same margins as P , since

$$\sum_j q_{ij} = p_{i+}, \quad \sum_i q_{ij} = p_{+j}. \quad (\text{E.3})$$

The factorisation in Equation (E.1) makes X and Y independent under Q , so $I(Q) = 0$.

The divergence from P to this reference is

$$D_{\text{KL}}(P||Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}} \quad (\text{E.4})$$

$$= \sum_{i,j} p_{ij} \log \frac{p_{ij}}{p_{i+}p_{+j}} \quad (\text{E.5})$$

$$= I(P). \quad (\text{E.6})$$

Consequently, testing $I(P) = I(Q)$ for this constructed reference is exactly testing $I(P) = 0$, or independence.

E.2 Why the first-order term vanishes

At independence, $p_{ij} = p_{i+}p_{+j}$, and every population pointwise-MI value is zero:

$$\ell_P(i, j) = \log 1 = 0. \quad (\text{E.7})$$

Equation (4.18) therefore gives

$$\left. \frac{dI(P_\varepsilon)}{d\varepsilon} \right|_0 = \ell_P(x, y) - I(P) = 0. \quad (\text{E.8})$$

The first-order variance also vanishes:

$$V(P) = \text{Var}_P\{\ell_P(X, Y)\} = 0. \quad (\text{E.9})$$

The denominator used by Normal Wald and Expanded Welch is thus degenerate under the population independence null. This does not mean that a finite sample must have zero plug-in MI or zero plug-in variance. Sampling noise usually makes the empirical table depart from exact independence. The point is that these random departures do not enter MI through the usual first-order term at the null. They enter through the second-order term derived next, so changing only the degrees of freedom of a first-order statistic cannot supply the missing source of variation.

E.3 Second-order behaviour

Fix a strictly positive independent table P , and let $h = (h_{ij})$ describe a small redistribution of probability, with $\sum_{i,j} h_{ij} = 0$. Define the path

$$P_\varepsilon = P + \varepsilon h, \quad f(\varepsilon) = I(P_\varepsilon), \quad (\text{E.10})$$

for values of ε small enough that all cells remain positive. Here $\varepsilon = 0$ is the expansion point and $f(0) = I(P) = 0$. The second-order Taylor expansion is

$$f(\varepsilon) \approx \underbrace{f(0)}_{\text{zeroth order}} + \underbrace{f'(0)\varepsilon}_{\text{first order}} + \underbrace{\frac{f''(0)}{2!}\varepsilon^2}_{\text{second order}}. \quad (\text{E.11})$$

The first-order term is

$$f'(0) = \sum_{i,j} \{\ell_P(i,j) - 1\} h_{ij} = 0. \quad (\text{E.12})$$

Both parts vanish: $\ell_P(i,j) = 0$ at independence, and the probability constraint gives $\sum_{i,j} h_{ij} = 0$. Thus, positive and negative cell errors cannot create a linear change in MI around the independence table.

To state the surviving term, define the row and column changes $h_{i+} = \sum_j h_{ij}$ and $h_{+j} = \sum_i h_{ij}$. Differentiating the entropy form of MI twice gives

$$f''(0) = \sum_{i,j} \frac{h_{ij}^2}{p_{ij}} - \sum_i \frac{h_{i+}^2}{p_{i+}} - \sum_j \frac{h_{+j}^2}{p_{+j}}. \quad (\text{E.13})$$

Because $p_{ij} = p_{i+}p_{+j}$, define the part of each cell change that cannot be explained by changed margins as

$$a_{ij} = h_{ij} - p_{+j}h_{i+} - p_{i+}h_{+j}. \quad (\text{E.14})$$

Equation (E.13) can then be written as

$$f''(0) = \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}. \quad (\text{E.15})$$

Substituting the zeroth-, first-, and second-order terms into Equation (E.11) gives

$$I(P_\varepsilon) = \underbrace{0}_{\text{zeroth and first order}} + \underbrace{\frac{\varepsilon^2}{2} \sum_{i,j} \frac{a_{ij}^2}{p_{i+}p_{+j}}}_{\text{second order}} + o(\varepsilon^2). \quad (\text{E.16})$$

The remainder $o(\varepsilon^2)$ is negligible relative to ε^2 as ε approaches zero.

For a multinomial sample, take $\varepsilon h = \hat{P} - P$. An empirical cell proportion has standard deviation $\sqrt{p_{ij}(1-p_{ij})/n}$, so each εh_{ij} is typically proportional to $n^{-1/2}$. The corresponding row- and column-adjusted errors εa_{ij} have the same scale. Equation (E.16) squares these errors, making $\hat{I}(P)$ proportional to n^{-1} under independence. Multiplying by $2n$ therefore produces a statistic with a non-vanishing scale:

$$2n\hat{I}(P) \xrightarrow{d} \chi_{(r-1)(c-1)}^2. \quad (\text{E.17})$$

This is the usual likelihood-ratio, or G , test of independence under a fixed alphabet and the standard positive-support large-sample conditions (Wilks, 1938; Brillinger, 2004). The chi-squared reference describes the squared cell errors in Equation (E.16).

Expanded Welch instead standardises a signed first-order MI error on the $n^{-1/2}$ scale. At independence that signed term is absent, so its Student reference is not the chi-squared reference in another form.

Although squaring a regular asymptotic $N(0, 1)$ statistic gives a χ_1^2 limit, the present T is degenerate at independence. Its square is therefore not the likelihood-ratio G statistic.

The same likelihood-ratio connection appears for conditional information measures. For finite Markov chains, [Barnett and Bossomaier \(2012\)](#) relate transfer entropy to a log-likelihood ratio with an asymptotic chi-squared reference; the wider information-theoretic setting is reviewed by [Bossomaier et al. \(2016\)](#). This is an adjacent application of the same second-order likelihood principle, not a two-sample equal-MI Welch test.

E.4 What alternative constructions produce

Using the observed margins to set $\hat{q}_{ij} = \hat{p}_{i+}\hat{p}_{+j}$ gives

$$2nD_{\text{KL}}(\hat{P}||\hat{Q}) = 2n\hat{I}(P), \quad (\text{E.18})$$

which is the likelihood-ratio statistic above. Because $\hat{Q}$ is constructed from the same sample as $\hat{P}$, it is not an independent second table and cannot be inserted into the two-sample Welch variance formula as though it supplied new information.

Repeatedly shuffling one variable preserves its observations while breaking the pairing with the other variable. The resulting empirical distribution is a permutation reference. Repeatedly drawing tables from $\hat{P}_X\hat{P}_Y$ is a parametric bootstrap reference. A single simulated Q table adds Monte Carlo variation but does not by itself define a calibrated null distribution.

The independent reference $Q = P_X P_Y$ is therefore mathematically valid, but it does not create a regular two-sample MI comparison. Analytic improvement of the sparse independence test would require a second-order finite-sample method and is not an immediate extension of the Expanded Welch calculation.

Appendix F

Reproducibility and Protocol Record

The final protocol is `WelchSatterthwaiteMI/experiments/THESIS_REDESIGN_PROTOCOL.json`, version 1.0.0, frozen on 7 September 2026 UTC before the final simulation. It specifies the population grids, seeds, methods, 20,000 replicates, $\alpha = 0.05$, validation tolerances and timing design. This local freeze followed exploratory development; it was not an external preregistration.

The run contains 3,111 unique configurations, 4,001 display positions and 534 population pairs, corresponding to 62,220,000 sampled table pairs. The saved verification records check the population targets, result denominators, paired method outputs, validity rules and supplementary results report. Chapter 6 uses the same frozen tables; no population was resimulated for a thesis figure.

The software requires matching table shapes, at least two rows and columns, finite nonnegative integer counts, and more than one observation in each table. Wald requires a finite corrected difference and standard error, $\hat{V}(P) + \hat{V}(Q) > 10^{-14}$, and a finite p-value. Expanded Welch also requires positive finite component and combined degrees of freedom and a finite p-value. It does not calculate a combined degree of freedom when a component degenerates, even if the direct combined expression in Appendix A has a value. Failed conditions are recorded as invalid calculations.

The exact software versions, seeds and source hashes are recorded in `WelchSatterthwaiteMI/results/thesis_redesign/run_metadata.json`. The experiment supplement contains the protocol, implementation, consolidated results, 3,111 configuration checkpoints, supplementary results report, mechanism diagnostics and tests. Its README records minor post-run runner changes that leave the frozen statistical calculations unchanged.

The evidence audit reconstructs the protocol manifests and checks every figure and Chapter 6 claim against the saved files. Final checksums are in `SHA256SUMS.txt`. After extracting the source and supplement into one archive root, run these commands from the directory containing `main.tex` and `WelchSatterthwaiteMI/`:

```
python figures_rewrite/make_figures.py
python figures_rewrite/audit_evidence.py
latexmk -pdf main.tex
```

Bibliography

- Lionel Barnett and Terry Bossomaier. Transfer entropy as a log-likelihood ratio. *Physical Review Letters*, 109(13):138105, 2012. doi: 10.1103/PhysRevLett.109.138105.
- Robert M. Bell and Daniel F. McCaffrey. Bias reduction in standard errors for linear regression with multi-stage samples. *Survey Methodology*, 28(2):169–181, 2002. URL <https://www150.statcan.gc.ca/n1/pub/12-001-x/2002002/article/9058-eng.pdf>.
- Thomas B. Berrett and Richard J. Samworth. USP: An independence test that improves on Pearson’s chi-squared and the G -test. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 477(2256):20210549, 2021. doi: 10.1098/rspa.2021.0549.
- Terry Bossomaier, Lionel Barnett, Michael Harré, and Joseph T. Lizier. *An Introduction to Transfer Entropy: Information Flow in Complex Systems*. Springer, Cham, 2016. doi: 10.1007/978-3-319-43222-9.
- Richard P. Brent. *Algorithms for Minimization without Derivatives*. Prentice-Hall, Englewood Cliffs, New Jersey, 1973.
- David R. Brillinger. Some data analyses using mutual information. *Brazilian Journal of Probability and Statistics*, 18(2):163–183, 2004. URL <https://www.stat.berkeley.edu/~brill/Papers/MIBJPS.pdf>.
- EunYi Chung and Joseph P. Romano. Exact and asymptotically robust permutation tests. *Annals of Statistics*, 41(2):484–507, 2013. doi: 10.1214/13-AOS1090.
- W. Edwards Deming and Frederick F. Stephan. On a least squares adjustment of a sampled frequency table when the expected marginal totals are known. *Annals of Mathematical Statistics*, 11(4):427–444, 1940. doi: 10.1214/aoms/1177731829.
- Robert Drake and Apratim Guha. A mutual information-based k -sample test for

- discrete distributions. *Journal of Applied Statistics*, 41(9):2011–2027, 2014. doi: 10.1080/02664763.2014.899325.
- Bradley Efron. Bootstrap methods: Another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979. doi: 10.1214/aos/1176344552.
- Michael P. Fay and Barry I. Graubard. Small-sample adjustments for Wald-type tests using sandwich estimators. *Biometrics*, 57(4):1198–1206, 2001. doi: 10.1111/j.0006-341X.2001.01198.x.
- Apratim Guha and Tom Chothia. A two sample test based on mutual information. *Calcutta Statistical Association Bulletin*, 66(1–2):39–54, 2014. doi: 10.1177/0008068320140103.
- Frank R. Hampel. The influence curve and its role in robust estimation. *Journal of the American Statistical Association*, 69(346):383–393, 1974. doi: 10.1080/01621459.1974.10482962.
- Kermit Hutcheson. A test for comparing diversities based on the Shannon formula. *Journal of Theoretical Biology*, 29(1):151–154, 1970. doi: 10.1016/0022-5193(70)90124-4.
- Marcus Hutter. Distribution of mutual information. In *Advances in Neural Information Processing Systems 14*, pages 399–406, 2002. Conference held in 2001; arXiv:cs/0112019.
- Marcus Hutter and Marco Zaffalon. Distribution of mutual information from complete and incomplete data. *Computational Statistics & Data Analysis*, 48(3): 633–657, 2005. doi: 10.1016/j.csda.2004.03.010.
- Guido W. Imbens and Michal Kolesár. Robust standard errors in small samples: Some practical advice. *The Review of Economics and Statistics*, 98(4):701–712, 2016. doi: 10.1162/REST_a_00552.
- Arnold Janssen. Studentized permutation tests for non-i.i.d. hypotheses and the generalized Behrens–Fisher problem. *Statistics & Probability Letters*, 36(1):9–21, 1997. doi: 10.1016/S0167-7152(97)00043-6.
- Kirthivasan Kandasamy, Akshay Krishnamurthy, Barnabas Poczos, Larry Wasserman, and James M. Robins. Nonparametric von Mises estimators for entropies, divergences and mutual informations. In *Advances in Neural Information Processing Systems 28*, pages 397–405, 2015.

- Göran Kauermann and Raymond J. Carroll. A note on the efficiency of sandwich covariance matrix estimation. *Journal of the American Statistical Association*, 96(456):1387–1396, 2001. doi: 10.1198/016214501753382309.
- Marius Marinescu and Costel Balcau. On the use of mutual information for testing independence, 2025. arXiv:2502.17636.
- George A. Miller. Note on the bias of information estimates. In Henry Quastler, editor, *Information Theory in Psychology: Problems and Methods*, pages 95–100. Free Press, Glencoe, Illinois, 1955.
- Roelof Moddemeijer. On estimation of entropy and mutual information of continuous distributions. *Signal Processing*, 16(3):233–248, 1989. doi: 10.1016/0165-1684(89)90132-1.
- Roelof Moddemeijer. A statistic to estimate the variance of the histogram-based mutual information estimator based on dependent pairs of observations. *Signal Processing*, 75(1):51–63, 1999. doi: 10.1016/S0165-1684(98)00224-2.
- Ricardo Mora and Javier Ruiz-Castillo. The statistical properties of the mutual information index of multigroup segregation. Working Paper, Economics Series 48 09-84, Universidad Carlos III de Madrid, Departamento de Economía, 2009.
- Ricardo Mora and Javier Ruiz-Castillo. Entropy-based segregation indices. *Sociological Methodology*, 41(1):159–194, 2011. doi: 10.1111/j.1467-9531.2011.01237.x.
- Wei Pan and Melanie M. Wall. Small-sample adjustments in using the sandwich variance estimator in generalized estimating equations. *Statistics in Medicine*, 21(10):1429–1441, 2002. doi: 10.1002/sim.1142.
- Liam Paninski. Estimation of entropy and mutual information. *Neural Computation*, 15(6):1191–1253, 2003. doi: 10.1162/089976603321780272.
- Mark S. Roulston. Estimating the errors on measured entropy and mutual information. *Physica D: Nonlinear Phenomena*, 125(3–4):285–294, 1999. doi: 10.1016/S0167-2789(98)00269-3.
- Franklin E. Satterthwaite. An approximate distribution of estimates of variance components. *Biometrics Bulletin*, 2(6):110–114, 1946. doi: 10.2307/3002019.
- Claude E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423, 623–656, 1948. Parts I and II.

- Aad W. van der Vaart. *Asymptotic Statistics*. Cambridge University Press, Cambridge, 1998. doi: 10.1017/CBO9780511802256.
- Abraham Wald. Tests of statistical hypotheses concerning several parameters when the number of observations is large. *Transactions of the American Mathematical Society*, 54(3):426–482, 1943. doi: 10.1090/S0002-9947-1943-0012401-3.
- Bernard L. Welch. The generalization of “Student’s” problem when several different population variances are involved. *Biometrika*, 34(1–2):28–35, 1947. doi: 10.1093/biomet/34.1-2.28.
- Samuel S. Wilks. The large-sample distribution of the likelihood ratio for testing composite hypotheses. *The Annals of Mathematical Statistics*, 9(1):60–62, 1938. doi: 10.1214/aoms/1177732360.
- Edwin B. Wilson. Probable inference, the law of succession, and statistical inference. *Journal of the American Statistical Association*, 22(158):209–212, 1927. doi: 10.1080/01621459.1927.10502953.